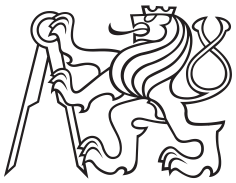
Master Thesis



Czech
Technical
University
in Prague

F3

Faculty of Electrical Engineering
Department of Control Engineering

# Vehicle drift maneuver stabilization

Bc. Jan Kohout

Supervisor: doc. Ing. Tomáš Haniš, Ph.D.
Field of study: Cybernetics and Robotics
May 2025

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Master’s thesis title in English:

**Vehicle drift maneuver stabilization**

Master’s thesis title in Czech:

**Stabilizace manévru vozu s vysokým stranovým skluzem**

Guidelines:

The primary objective of this thesis is to implement control algorithms for high-dynamic lateral maneuvers, specifically within the drift and power-slide domain. Both over-actuated vehicles with rear-axle steering and conventional platforms will be considered.
1. Develop and analyze mathematical models for both conventional and over-actuated platforms. Study high side-slip maneuvers (at vehicle CoG) such as drifting and power sliding.
2. Propose a traction control algorithm architecture to regulate vehicle lateral dynamics for these platforms and maneuvers.
3. Implement the proposed algorithms for both conventional and over-actuated platforms.
4. Evaluate the performance of the implemented algorithms.

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Finally, I want to express heartfelt thanks to my girlfriend for her endless support, love, and patience. Her optimism and emotional support have helped me stay focused and persistent even during long nights and setbacks.


# Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 10. May 2025

# Abstract

This thesis addresses the problem of stabilizing vehicles during high side-slip angle maneuvers, which are characteristic of drifting. The work compares the performance of a classical two-wheel steering (2WS) platform and an overactuated four-wheel steering (4WS) platform. A nonlinear single-track vehicle model was implemented, and Linear Quadratic Regulator (LQR) and PI controllers are designed for both configurations. Simulation scenarios include a highway lane-change maneuver and a tight U-turn. Results show that 4WS significantly enhances stability, maneuverability, and control authority during high side-slip motion. The platform also demonstrates reduced lap times and potential for safer and more efficient autonomous driving at the limit of handling. The influence of 4WS on vehicle performance, power consumption, and tire load utilization is analyzed and discussed.



**Keywords:** vehicle dynamics, drifting, four-wheel steering, LQR, linear control, modeling, PID



**Supervisor:** doc. Ing. Tomáš Haniš, Ph.D.



# Abstrakt

Tato diplomová práce se zabývá stabilizací vozidla při manévrech s vysokým stranovým skluzem, typických pro řízené smyky (drifting). Je zde porovnávána klasická konfigurace s řízením předních kol (2WS) s přeaktuovanou platformou se čtyřkolovým řízením (4WS). Byl implementován nelineární model vozidla s jednou stopou, pro který byly navrženy regulátory typu LQR a PI. Simulační scénáře zahrnují změnu jízdního pruhu na dálnici a ostrou zatáčku ve tvaru U. Výsledky ukazují, že 4WS výrazně zlepšuje stabilitu, manévrovatelnost a ovladatelnost vozidla při pohybu s vysokým stranovým skluzem. Tato platforma navíc dosahuje kratších časů na kolo a vykazuje potenciál pro bezpečnější a efektivnější autonomní řízení na hranici adheze. V práci je analyzován vliv 4WS na jízdní vlastnosti, spotřebu energie a využití adhezních sil pneumatik.



**Klíčová slova:** dynamika vozidla, drifting, čtyřkolové řízení, LQR, linarní řízení, modelování, PID

**Překlad názvu:** Stabilizace manévru vozu s vysokým stranovým skluzem

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# List of Abbreviations

**ABS** Anti-lock Braking System

**CG** Center of Gravity

**ESP** Electronic Stability Program

**FWD** Front-Wheel Drive

**RWD** Rear-Wheel Drive

**2WS** Two-Wheel Steering

**4WS** Four-Wheel Steering

**LQR** Linear Quadratic Regulator

**MPC** Model Predictive Control

**PI** Proportional-Integral Controller

**PID** Proportional-Integral-Derivative Controller

**MLTP** Minimum Lap Time Problem

**NMPC** Nonlinear Model Predictive Control

$v_x$ Longitudinal Velocity

$v_y$ Lateral Velocity

$\psi$ Yaw Angle

$\dot{\psi}$ Yaw Rate

$\delta_f$ Front Wheel Steering Angle

$\delta_r$ Rear Wheel Steering Angle

$\alpha_f$ Front Slip Angle

$\alpha_r$ Rear Slip Angle

$\lambda$ Slip Ratio

$F_x$ Longitudinal Force

$F_y$ Lateral Force

$F_z$ Vertical Force

**CAD** Computer-Aided Design

**RC** Radio-Controlled

# Chapter 1

# Introduction

As automotive technology continues to evolve, innovation remains a driving force behind the development of modern vehicles. Today's cars are equipped with an extensive array of advanced sensors and actuators, enabling the implementation of increasingly sophisticated control algorithms. Fundamental safety systems such as Anti-lock Braking Systems (ABS) and Electronic Stability Programs (ESP) form the foundation upon which more complex driver assistance features are built. These include adaptive cruise control, lane-keeping assistance, and automatic emergency braking.

Many manufacturers are now aiming even higher—towards delivering a fully autonomous driving experience. As computational capabilities grow, vehicles are no longer constrained by traditional mechanical architectures limited to a steering wheel and three pedals. This opens up new design opportunities and control strategies.

One particularly compelling concept is the use of four-wheel steering (4WS) shown in figure 1.1 By introducing an additional degree of freedom, RWS allows for enhanced vehicle maneuverability and stability, especially during dynamic driving conditions. Though this idea dates back to the latter half of the 20th century, its real-world implementation was limited to a few high-end or experimental vehicles. With recent advances in control systems and actuation technology, interest in RWS is being rekindled.

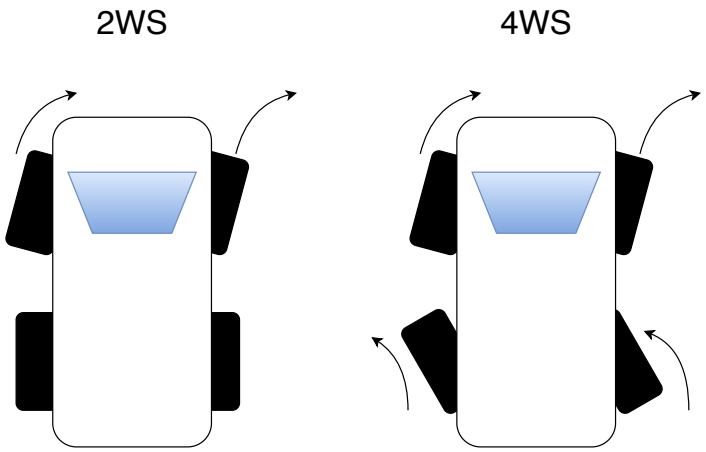



**Figure 1.1:** Illustration of tow wheel steering (2WS) and four wheel steering (4WS) platforms.

This thesis explores the control of vehicles undergoing high side-slip angle in a way that is not associated with aggressive or drifting-like behavior. The study compares the performance of two vehicle configurations: the conventional front-wheel steering (2WS) layout and an overactuated platform with four-wheel steering (4WS). The objective is to analyze how rear-wheel steering can enhance vehicle stability and controllability in these challenging scenarios.

## 1.1 Motivation

The motivation for this thesis primarily stems from my passion for cars and motorsports. The following question was asked

What benefits does drifting offer to the end customer?

I can start answering by focusing on the safety aspect. As mentioned previously, modern cars have control systems to prevent oversteer. However there are real-life situations where a car cannot be stabilized and consequently enters the unstable oversteering area of driving. In this case having a system that stabilizes the oversteer into a drift will improve safety. Adding possibility to drift can improve autonomous driving algorithms as it enlarges the set of possible maneuvers. One example is a lane change maneuver on highway. This enables more autonomous driving scenarios to be possible, meaning the car does not have to hand over the control to the driver. This will improve

reaction times in dangerous situations.

Additionaly, the aspect of increased maneuverability can be considered. As using high side-slip angle maneuver may be used to drive around obstacles more precisely and with higher levels of steering control. Parking in tight spots can be done with more precision and less risk of damaging the car.

To stabilize the car during oversteer Drifting is a key skill for any racing driver. Its most pronounced application is in rally sports, where many drivers can be seen employing drifting through tight corners as it can increase their exit speed and subsequently increase their lap time. The discipline of drifting has also become a sport in its own right, with competitions and events dedicated to showcasing the skill and control required to perform controlled slides.

Recreational use of high side-slip maneuvers should also be considered as an important aspect. For many car manufacturers, adding drift stabilization can be used as a marketing tool. Drifting is seen as a dangerous, high-level skill activity done for fun and in defiance to norm.

Last but not least, there is the question of how the 4WS can improve classical drifting.

Drifting with two-wheel steering platform is known to be very fuel inefficient and also associated with high tire wear. Incorporating four-wheel steering enables more efficient, precise, and rapid initiation and control of the drift maneuver, enhancing the ecology aspect of drifting. Four-wheel steering can be used to reduce the tire wear and fuel consumption as well as overall wear of the drive train.

## 1.2 Hypothesis

Main goal of this thesis is to prove the following:

- A control strategy and a controller can be developed to stabilize the car in high side-slip angle maneuvers for two-wheel steering and four-wheel steering platform.
- There are comparison variables that can be drawn between the two platforms.
- The four-wheel steering can be used to improve the performance of the car in high side-slip angle manevours.
- The four-wheel steering can be used to reduce the tire wear and increase safety in these manevours.

## 1.3 State of the art

### 1.3.1 Mathematical models

Modeling of car dynamics and tire to road surface is a well studied topic. In [28], the authors present methodically a variety of models ranging from simple kinematic single-track models to complex dynamic twin-rack models, including drivetrain, wheel and wind resistance models. Modeling is done using ordinary differential equations. Chapter discussing different tire models ranging from mathematical to physical models is also present.

The author of [25] presents overview of control sytems for lateral and longitudinal dynamics. Topics like vehicle stability control, adaptive cruise control and lane keeping are discussed. Single-track and twin-track models with fluctuating complexity are discussed.

The book [23] is one of the most well known works on tire modeling. Mutiple tire models are presented. The well-known Pacejka magic formula is presented and described in detail. Less known and used models like stretched string model and brush model are also presented.

### 1.3.2 Two-wheel steering platform

Ono et al. [21] investigated the bifurcation behavior in vehicle dynamics and proposed a robust front wheel steering control method based on $\mathcal{H}_\infty$ synthesis. Their analysis showed that saddle-node bifurcations, caused by rear tire force saturation, can lead to vehicle spin. They modeled the saturation as an uncertain nonlinearity and designed a controller that ensures robust stability and improved cornering performance, resembling the counter-steering technique used by skilled drivers.

Building upon this, Hindiyeh and Gerdes [12] analyzed drift equilibria using a three-state rear-wheel-drive bicycle model that incorporates longitudinal dynamics and lateral-longitudinal force coupling. Their work demonstrated the existence and saddle-type instability of drift equilibria, providing insight into vehicle behavior beyond the limits of handling.

Velenis et al. [31, 32] explore advanced drift control techniques for both front-wheel-drive (FWD) and rear-wheel-drive (RWD) vehicles. In the case of FWD vehicles, the authors analyze the handbrake-cornering technique using experimental data from a rally driver and derive control inputs that stabilize the vehicle during high side-slip maneuvers. For RWD vehicles, they propose a control scheme based on coordinated steering and drive torque inputs to stabilize drifting equilibria. Their approach mimics expert driver behavior and is implemented on a comprehensive four-wheel vehicle model. Simulation results validated with experimental data confirm the effectiveness of these strategies in achieving controlled drifting even on low-radius paths, significantly expanding the mobility capabilities of both FWD and RWD platforms.

Recent work has expanded the experimental and theoretical understanding of autonomous drifting. Park and Kang [24] proposed a drift controller using LQR to stabilize a 1:10 scale car on circular trajectories, validating its performance in real-time onboard experiments. Liniger [17] developed a nonlinear model predictive controller capable of controlling an RC car beyond its handling limits, demonstrating real-time control using both PID and receding horizon strategies. Song et al. [30] introduced an ILQR-based controller for steady-state drifting with analytically derived safety boundaries, showing how state constraints improve robustness. Edelmann and Plöchl [6] analyzed the powerslide regime using a nonlinear four-wheel model, characterizing stability and handling properties under high-slip, large-steering-angle conditions and validating their findings through real vehicle experiments.

The Stanford Dynamic Design Lab has extensively studied autonomous drifting, focusing on both theoretical modeling and experimental validation. Hindiyeh and Gerdes [11, 10] developed a controller framework based on a three-state rear-wheel-drive bicycle model that stabilizes unstable drift equilibria using coordinated steering and rear drive force. Goh and Gerdes [9] extended this approach to enable simultaneous path and sideslip tracking using MARTY, Stanford's autonomous drift car. More recently, Lenzo et al. [15] introduced torque vectoring with independent rear motors as a novel method to achieve energy-efficient autonomous drifting without braking. Finally, Weber and Gerdes [35] proposed a nonlinear model predictive control (NMPC) scheme capable of tracking dynamic, non-equilibrium drifting trajectories with high precision, validated on the full-scale vehicle Takumi.

### 1.3.3 Four-wheel steering platform

Yamamoto [36] introduced an active four-wheel steering (4WS) control system that combines a feedforward term based on steering wheel angle with a feedback term using yaw rate. This approach significantly improves transient response, reduces sideslip even during disturbance events, and enhances high-speed vehicle stability. Complementing this, Bolzern et al. [2] designed a robust full-active 4WS controller using $\mu$-synthesis techniques, targeting performance robustness against uncertainties in tire cornering stiffness. Their simulation results demonstrate strong rejection of disturbances such as side winds and split-μ braking, while maintaining low sideslip angles and high tracking performance.

Recent research has significantly advanced four-wheel steering (4WS) control strategies. Solea et al. [29] applied sliding-mode control to trajectory tracking of 4WS vehicles, demonstrating robustness under nonlinear and variable-speed conditions. Galvani et al. [8] compared 2WS and 4WS vehicles in emergency obstacle avoidance, showing that 4WS allows for later and safer evasive actions by exploiting both front and rear actuation. Lu et al. [18] introduced an adaptive LQR controller with GA-optimized parameters for 4WS electric vehicles, achieving improved stability and tracking accuracy across speeds. Rother et al. [26] developed a 1/10th scale JetRacer-4WS platform with model predictive control (MPC), providing an accessible physical testbed for advanced motion control algorithms. Lim et al. [16] conducted stability analysis for 4WS path tracking, revealing the benefits of tuning rear-to-front steering ratios depending on curvature and speed.

Recent developments in four-wheel steering (4WS) control have focused on improving robustness, adaptability, and coordination with other vehicle

dynamics control systems. Jia [13] proposed a robust nonlinear decoupling controller for 4WS vehicles that accounts for variations in vehicle mass, inertia, and road–tire interactions, enabling improved safety under varying speed and traction conditions. Zhang et al. [38] developed an optimal control strategy using yaw rate and centroid cornering angle feedback, which demonstrated superior trajectory tracking with no overshoot. Kang et al. [14] integrated front/rear torque control and braking into a layered 4WD EV controller, enhancing maneuverability and rollover prevention. Cao et al. [3] introduced an active rear steering system based on a time-varying 4WS reference model and model predictive control (MPC), which significantly improved yaw response and reduced driver input. Together, these works represent a shift toward comprehensive vehicle motion control frameworks where 4WS is integrated with adaptive, robust, or predictive methods.

## 1.4 Maneuver definition

### Oversteer

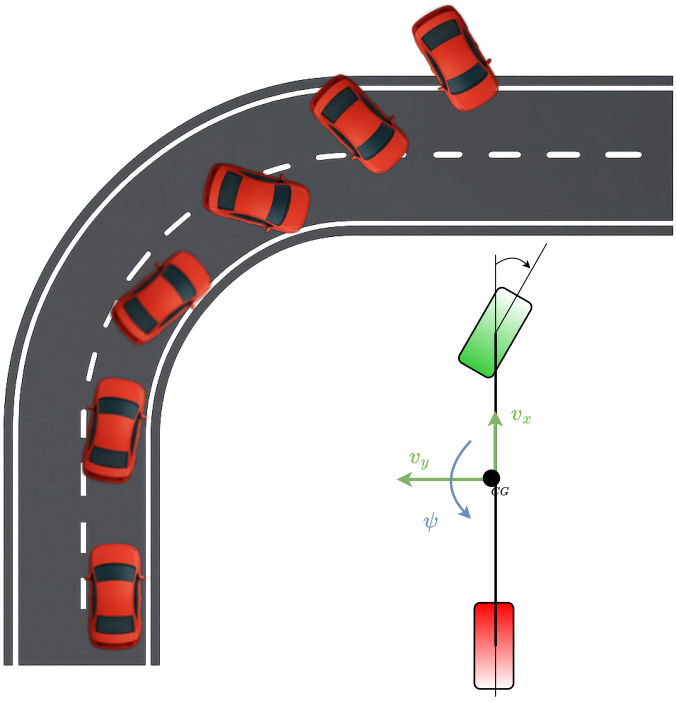



**Figure 1.2:** Illustration of oversteer dynamics during a maneuver.

Oversteer is a driving condition in which the rear wheels of vehicle lose

traction before the front wheels, causing the rear end to swing outwards during a turn. This can result in loss of control and may lead to a spin if not corrected. Designing cars to be oversteer-prone can enhance their agility and responsiveness, particularly in motorsport applications.

**Understeer**

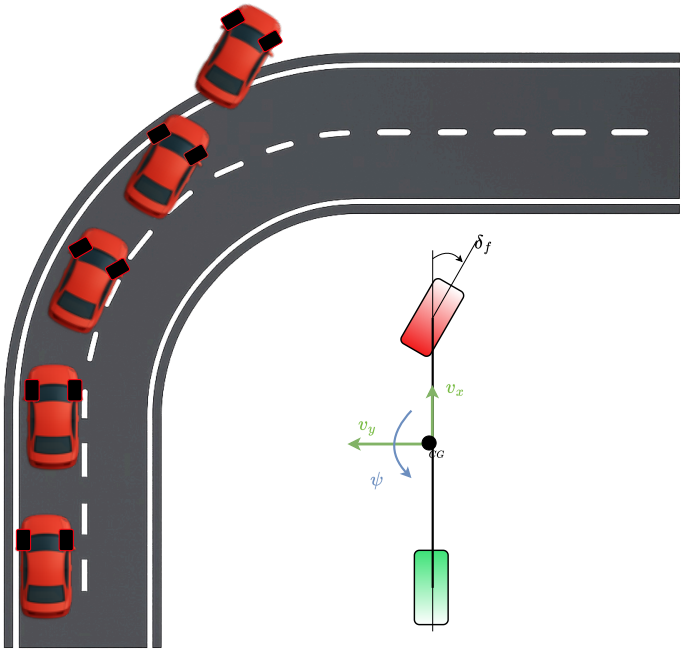



**Figure 1.3:** Illustration of understeer dynamics during a maneuver.

Understeer is a driving condition in which the front wheels of a vehicle lose traction before the rear wheels, causing the front end to push wide during a turn. This can result in a loss of control and may lead to a crash if not corrected.

**Drifting** is a controlled driving maneuver in which the driver intentionally induces and maintains oversteer, causing the rear wheels to lose traction through a corner. The vehicle is kept under control and follows a desired line using a combination of throttle, steering, clutch, handbrake, and braking inputs. During drift, the rear slip angle exceeds the front slip angle, and the car often faces different direction than its trajectory i.e. has high side-slip angle. Initiating the drift maenuver is done before apex of the corner. Usually, the driver forces rear slip ratio very close to minus one by using the handbreak, consequently overloading the rear tires. In combination with sharp steering input, the rear tires loose traction and the car starts to slide i.e rapid growth

of yaw rate. The driver then uses countersteering to stabilize the car in the drift.

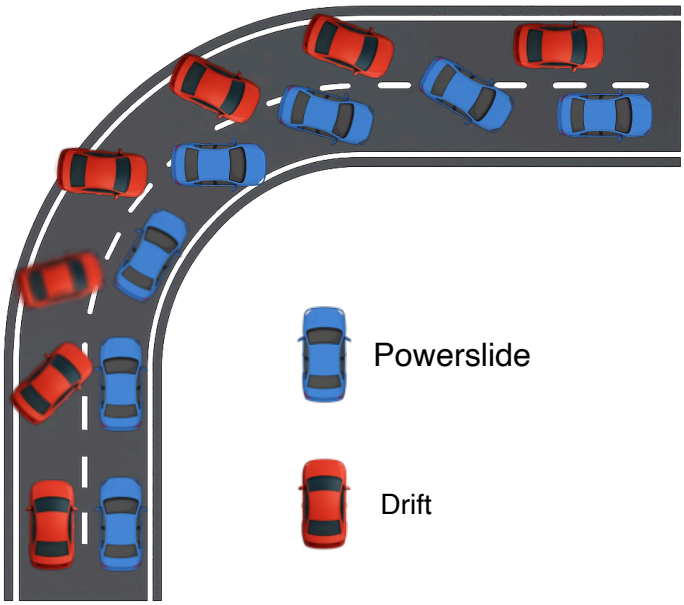



**Figure 1.4:** Illustration of powersliding and drifting maneuvers.

**Powerslide** refers to a transient condition in which a vehicle's rear wheels lose traction due to excessive longitudinal force. This results in the rear tires exceeding the available friction limit and operating near a slip ratio of 1, entering a saturated regime of the tire force curve. A powerslide is most commonly observed in rear-wheel-drive (RWD) vehicles and is generally initiated just after the apex of a corner, when the driver applies throttle to accelerate out of the turn. The defining characteristic of a powerslide is a sudden increase in yaw rate caused by oversteer, as the rear tires lose lateral grip while continuing to generate longitudinal force. Although powerslides can occur unintentionally due to driver error or abrupt torque application on low-friction surfaces, they are also deliberately exploited in motorsport disciplines such as drifting and rallying to maintain high cornering speeds or control vehicle orientation dynamically.

During both powerslide and drift, vehicle stability becomes highly dependent on precise modulation of throttle and steering inputs. If the rear slip angle exceeds a recoverable threshold and the driver is unable to countersteer appropriately, the vehicle may transition into a full spin. Real-world examples were studied in [31] and [32], where the autors publish caputed data from

profesional rally drivers.

As can be seen in figure 1.4, the main difference between drifting and powersliding is in the intiation of the maneuver. In drifting, the driver initiates the maneuver before the apex of the corner, while in powersliding, the driver initiates the maneuver after the apex of the corner. Powerslide is also associeted with the the loss of fricition on both axels thus sliding more than drifting. On the other hand the drift maneuver called reverse entry exists.

# Chapter 2

# High side-slip maneuver analysis

Laptime optimization is used by many racing teams to improve their performance. Solving minimum lap time problem (MLTP) is a very complex task as it is a non-linear, non-convex problem with many constraints. The minimum laptime problem was studied as early as 1958 [27], where the gear ratios and body was optimized for F1 track in Monza. In 1989 Metz and Williams [20] computed optimal control sequence using quasi-steady state optimization rutine. More detailed history of laptime optimization is decribed in [19]. Nowadays, MLTP is solved using different methods, like dynamic programming, genetic algorithms, nonlinear programming. Free MLTP programs like OptimumLap [22] can be obtained. MLTP toolbox exists for MATLAB [37] which formulates the problem as nonlinear program.

At CTU in Prague, the bachelor thesis of Richard Ciglanský [4] evaluates the impact of torque vectoring, four-wheel steering and ground effect production device on lap tim. It expands upon the optimisation framework developed by Ing. Martin Gurtner, Ph.D. The problem was defined using CasADi syntax [1], solved by the solver Ipopt [34].

The optimisation framework of Ing. Martin Gurtner, Ph.D. was used to simulate 2WS platform and 4WS platform. Both platforms were simulated on the same tracks with the same initail conditions and vehicle model parameters. The vehicle model used for following comparison is a single-track model.

## 2.1 Lane change simulation

In first scenario the car has to turn left go straight and than pull back. This simulates lanchange maneuver on the highway. Optimization toolbox was used to simulate both platofrms.

As can be seen in table 2.1, the 4WS platform was able to achieve lower lap time, higher average speed and also shorten the racing line length. This is supported by the state vector plots Resulting state vectors are shown in figure 2.1.The initail acceleration is the same for the boath platforms however the 4WS platform is able to sustain higher forward velocity. The lateral velocity of the 4WS platform is also higher. The 4WS platform reaches side-slip angle of 20 degrees while the 2WS platform reaches only 5 degrees.

| Track Parameter | 2WS | 4WS |
|---|---|---|
| Lap Time (s) | 3.57 | 3.22 |
| Average Speed (m/s) | 18.69 | 20.51 |
| Racing Line Length (m) | 66.72 | 66.03 |

**Table 2.1:** Comparison of Track Performance for 2WS and 4WS Platforms

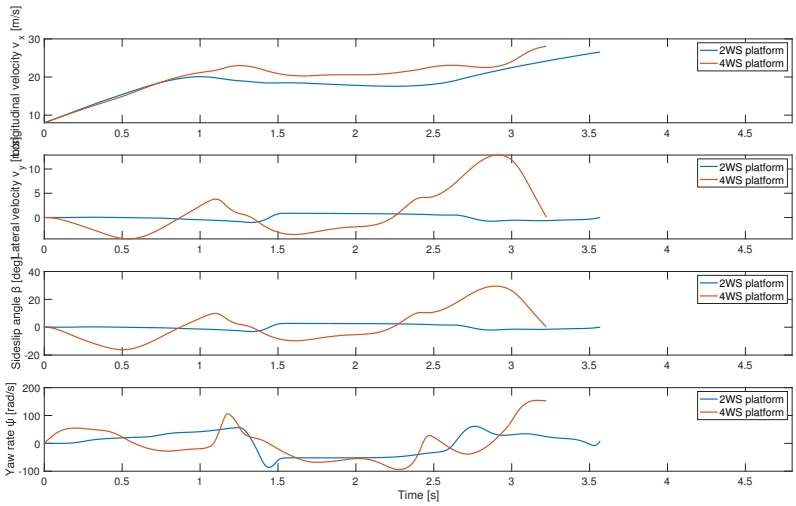



**Figure 2.1:** Vehicle state trajectories over time.

Resulting wheel speeds are shown in figure 2.2, here the 4WS has higher speeds all the time. Steering angle profiles are in figure 2.3. Here the Rear wheel steering angle was as high a as 30 degrees. The 4WS platform has over all higher steering angle.

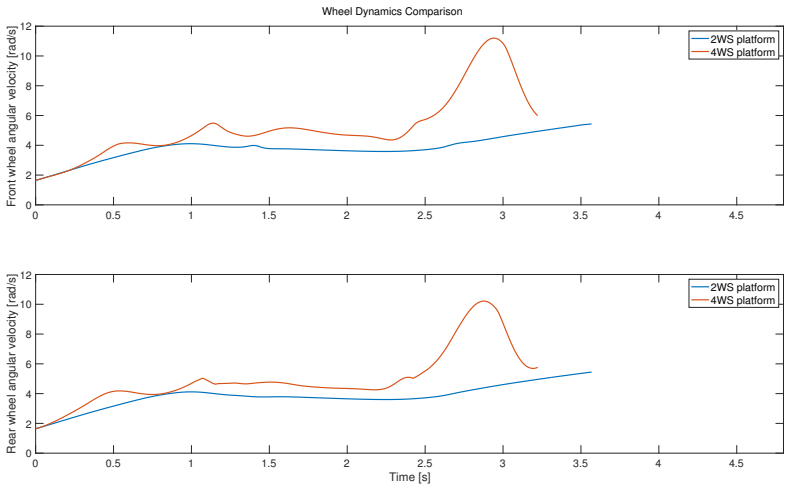



**Figure 2.2:** Wheel speeds during simulation.

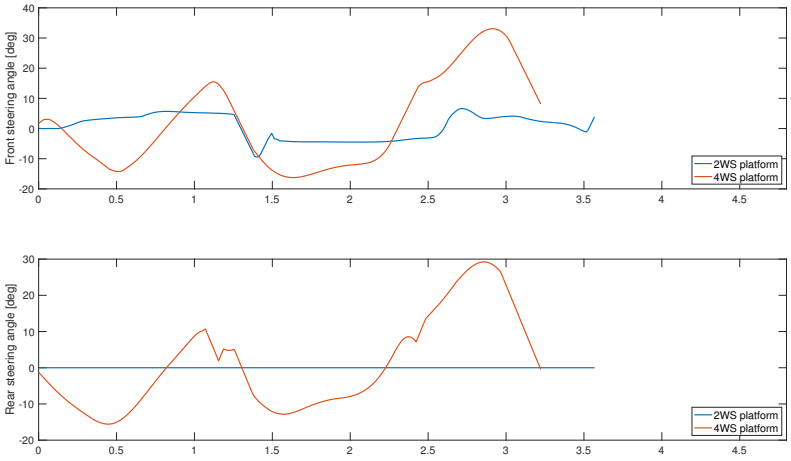



**Figure 2.3:** Steering angle profiles.

The racing lines for both platforms are shown in figure 2.4. The 4WS platform started to turn in much erlier. We can also see the higher side-slip angle of the 4WS platform.

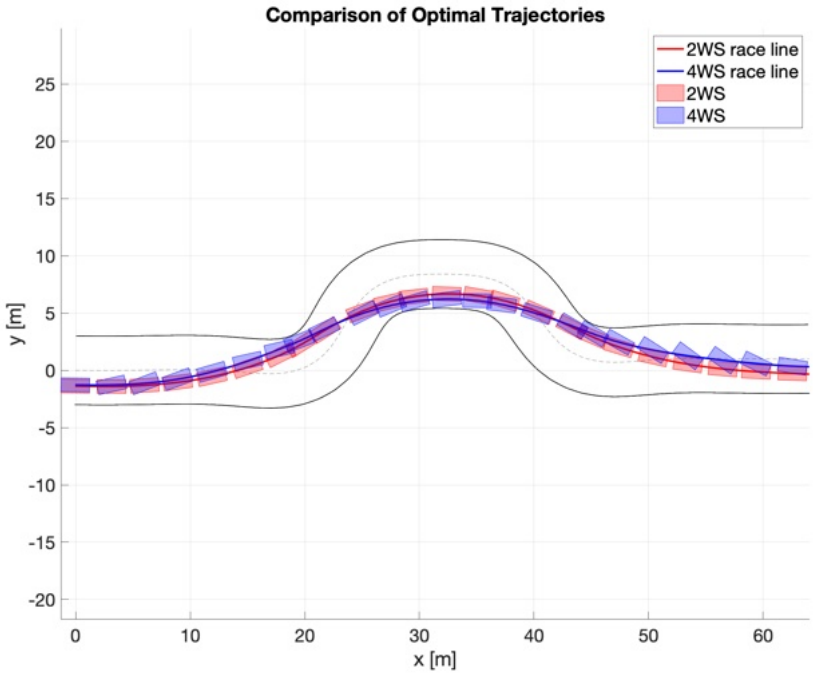



**Figure 2.4:** Race lines for both platforms on track.

## 2.2 U turn track

The second simulation was run on a track with a U-turn. The radius of the U-turn was set to 4 m. Same as previously, the starting and ending straights were set to 20 m. In table 2.2 We can see that the 4WS platform was able to achieve lower lap time, while maintaining higher average speed. The racing line was also shorter. The state vector plots in figure 2.5 show that the 4WS platform had higher lateral velocity and side-slip angle.

| Track Parameter | 2WS | 4WS |
|---|---|---|
| Lap Time (s) | 4.271 | 4.207 |
| Average Speed (m/s) | 12.23 | 12.42 |
| Racing Line Length (m) | 73.34 | 76.36 |

**Table 2.2:** Comparison of Track Performance for 2WS and 4WS Platforms

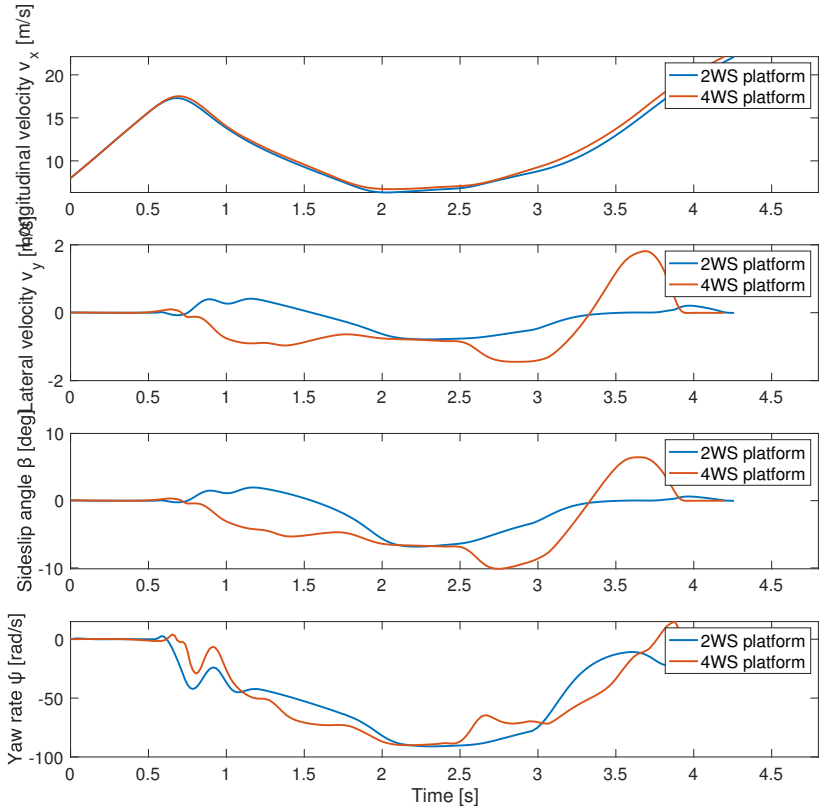



**Figure 2.5:** Vehicle state trajectories over time.

In figure 2.6, we can see that the front wheels of the 4WS platform were once again able to spin much faster because the lateral forces needed to be

generated were split by the front and rear wheels. In figure 2.7 we can see that the steering angles of 4WS platform were much higher. The 4WS platform performed coutersteering after third second of simulation, thus indicating a drift maneuver. The sideslip angle is around 10 degrees. This was not present for the 2WS platform.

In the figure 2.8 we can see the 4WS platform starts to turn into a corner later than the 2WS platform. This is due to the fact that the 4WS platform can generate lateral forces with the rear wheels. This allows the 4WS platform to turn into the corner later and with higher speed.

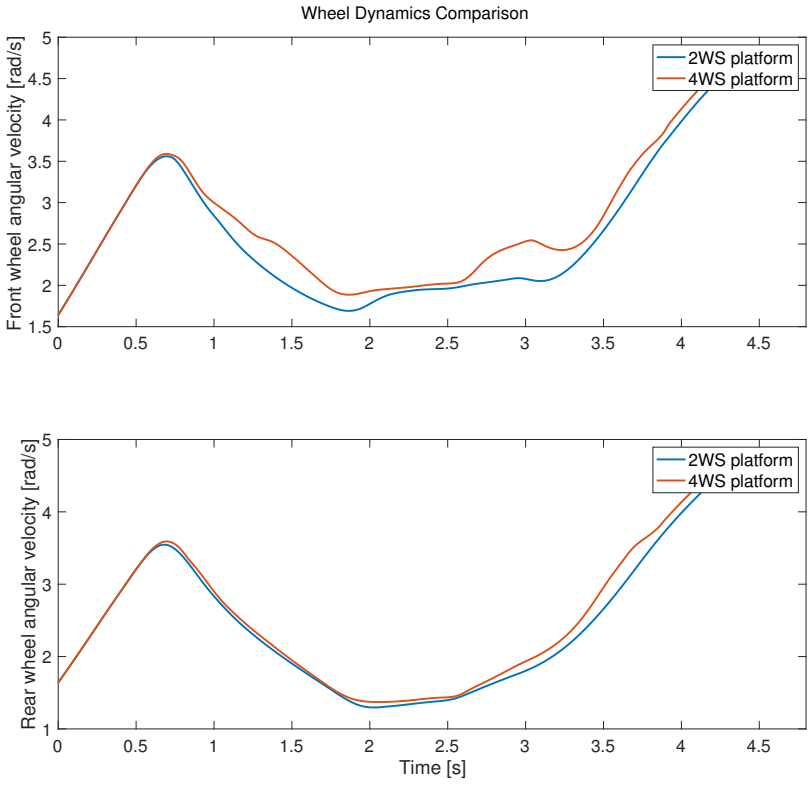



**Figure 2.6:** Wheel speeds during simulation.

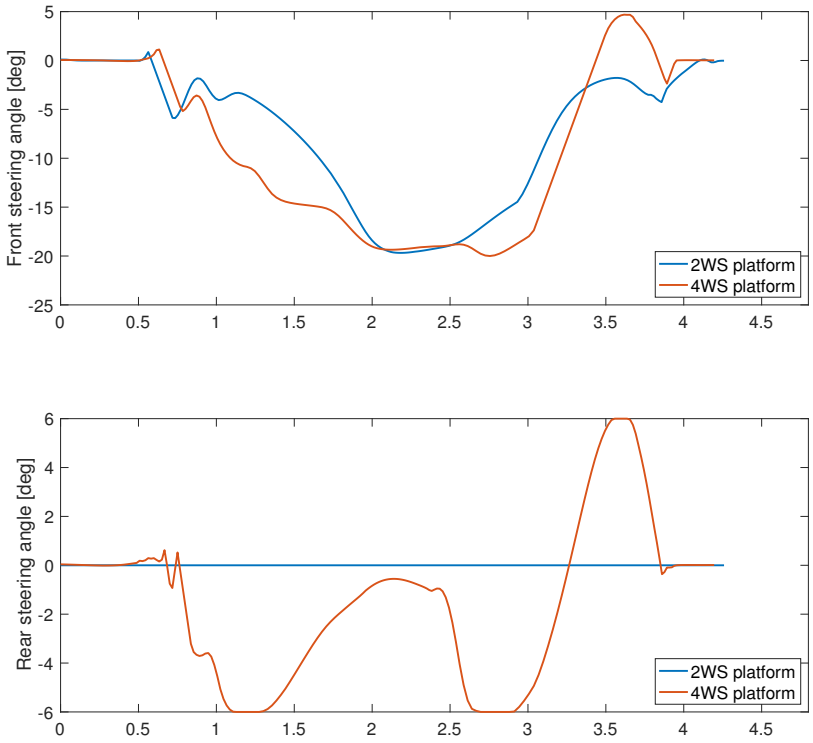



**Figure 2.7:** Steering angle profiles.

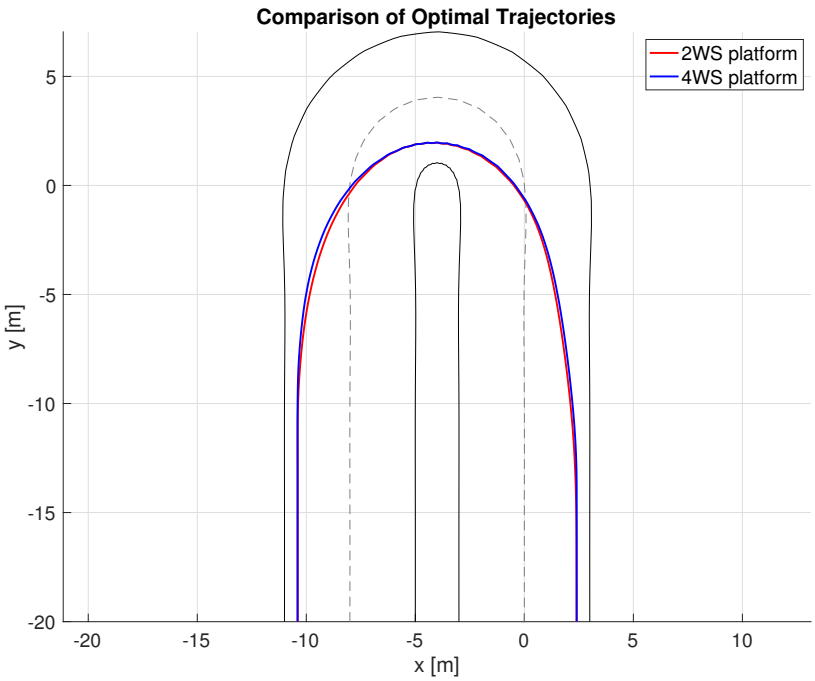



**Figure 2.8:** Race lines for both platforms on track.

# Chapter 3

# Mathematical models

In this chapter, different approaches to modeling car dynamics and tire-to-surface interaction are discussed. As usual, the choice of model represents a trade-off between complexity and accuracy. The chapter is split into two parts; first, the vehicle model is presented, and then the tire model is analyzed.

## 3.1 Vehicle model

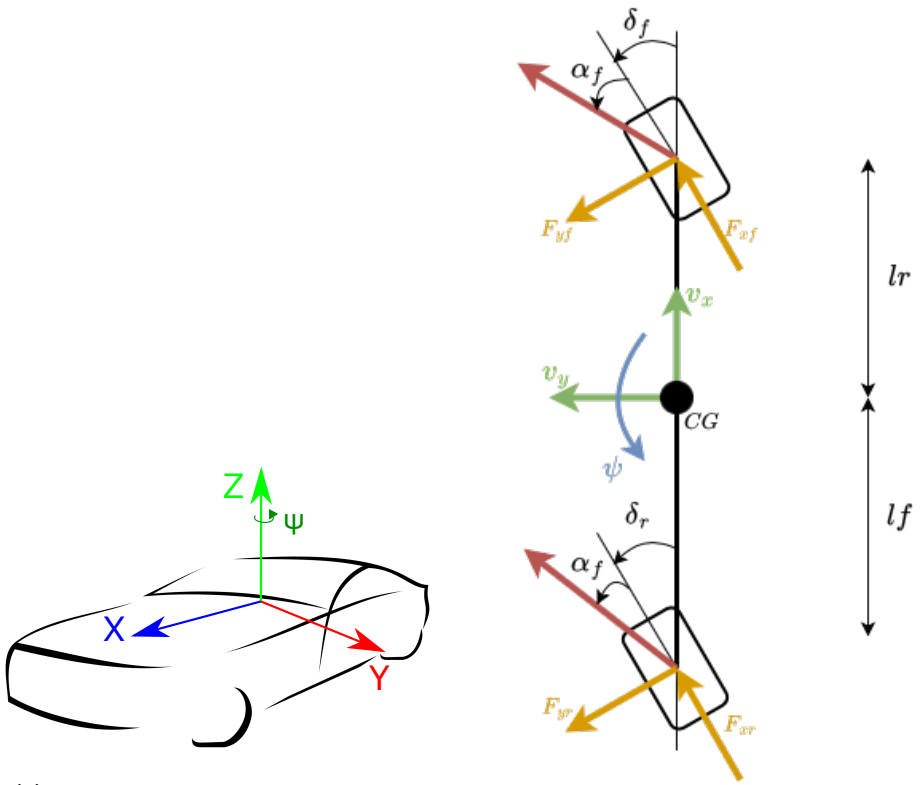
Generally speaking there are two approaches to modeling vehicle dynamics. The first one is the single-track model, which is a simplified model of a vehicle that represents it as a single track of wheels. The second one is the twin-track model, which represents the vehicle as two tracks of wheels, one for the front and one for the rear. Choosing the single-track model reduces the complexity quite substantially, without a meaningful loss of accuracy, which is why it was chosen for this experiment.

Following assumptions have been made:

- Rolling, pitching and lifting motion is neglected
- Vehicle mass is concentrated at the center of gravity
- Mass distribution is constant

- Contact point of tire lies along center of axel

### 3.1.1 Coordinate system and notation




**(a) :** The car coordinate system. Image from [7]

**(b) :** The single-track model notation.

**Figure 3.1:** Vehicle coordinate system and single-track model notation.

As can be seen in figure 3.1a the car coordinate system is a cartesian system with the x going forward, y going to the right and z going up. The origin is at the center of gravity (CG). Vehicle yaw angle is increasing in the counter clockwise direction.

In figure 3.1b the single track model notation is shown. Here we can see forces $F_x$ and $F_y$ acting on the center of gravity (CG) of the vehicle. $F_{x,i}$ and $F_{y,i}$ are forces acting upon the center of gravity of the $i$-th wheel along the x and y axes. $\beta$ is the side-slip angle. $\delta_i$ is the steering angle of the $i$-th wheel.

### 3.1.2 Model equations

To describe the car state there are two equally valid options. First uses vehicle velocity and side-slip angle and can be writen as follows:

$$\mathbf{x} = \begin{pmatrix} \beta \\ v \\ \dot{\psi} \end{pmatrix},$$

and the other uses lateral and logitudinal velocity like so:

$$\mathbf{x} = \begin{pmatrix} v_x \\ v_y \\ \dot{\psi} \end{pmatrix}.$$

Both descriptions are using the yaw rate $\dot{\psi}$. The following equations describe relationship between the two state vectors:

$$\begin{aligned} v &= \sqrt{v_x^2 + v_y^2}, \\ \beta &= \arctan\left(\frac{v_y}{v_x}\right). \end{aligned} \tag{3.1}$$

### 3.1.3 Model description using $\beta$, $v$ and $\psi$

Here the vehicle dynamics equations are taken from [7].

- Longitudinal motion

$$-mv(\dot{\beta} + \dot{\psi})\sin\beta + m\dot{v}\cos\beta = F_x \tag{3.2}$$

- Lateral motion

$$mv(\dot{\beta} + \dot{\psi})\cos\beta + m\dot{v}\sin\beta = F_y \tag{3.3}$$

- Yaw motion

$$I_z\ddot{\psi} = M_z, \tag{3.4}$$

where $m$ is the vehicle mass, $v$ is the velocity of CG, $\beta$ is the side-slip angle, $\psi$ is the yaw angle, $I_z$ is the moment of inertia around $z$ axis and $M_z$ is the yaw moment.

### 3.1.4 Model desciption using $vx$, $vy$ and $\psi$

Here the yaw motion is described the same way. The longitudinal and lateral motion are transformed using the equations 3.1

- Longitudinal force:
$$m(\dot{v}_x - \dot{\psi}\, v_y) = F_x,$$
- Lateral force:
$$m(\dot{v}_y + \dot{\psi}\, v_x) = F_y,$$
- Yaw moment:
$$I_z\, \ddot{\psi} = M_z.$$

Once again by rearranging the equations we can write the following:

$$\begin{aligned} \dot{v}_x &= \frac{1}{m}\big(F_x\big) - \dot{\psi}\, v_y, \\ \dot{v}_y &= \frac{1}{m}\big(F_y\big) + \dot{\psi}\, v_x, \\ \ddot{\psi} &= \frac{1}{I_z}\, M_z \end{aligned} \tag{3.5}$$

### 3.1.5 Center of gravity to body projection

The forces $F_x$ and $F_y$, as well as the moment $M_z$, result from the projection of steering angles. The transformation for four-wheel steering can be expressed in matrix form as:

$$\begin{pmatrix} F_x \\ F_y \\ M_z \end{pmatrix} = \begin{pmatrix} \cos\delta_f & -\sin\delta_f & \cos\delta_r & -\sin\delta_r \\ \sin\delta_f & \cos\delta_f & \sin\delta_r & \cos\delta_r \\ l_f\sin\delta_f & l_f\cos\delta_f & -l_r\sin\delta_r & -l_r\cos\delta_r \end{pmatrix} \begin{pmatrix} F_{xf} \\ F_{yf} \\ F_{xr} \\ F_{yr} \end{pmatrix} \tag{3.6}$$

In the case of conventional front-wheel steering only, the transformation matrix from equation (3.6) simplifies to:

$$\begin{pmatrix} F_x \\ F_y \\ M_z \end{pmatrix} = \begin{pmatrix} \cos\delta_f & -\sin\delta_f & 1 & 0 \\ \sin\delta_f & \cos\delta_f & 0 & 1 \\ l_f\sin\delta_f & l_f\cos\delta_f & 0 & -l_r \end{pmatrix} \begin{pmatrix} F_{xf} \\ F_{yf} \\ F_{xr} \\ F_{yr} \end{pmatrix} \tag{3.7}$$

The forces $F_{xf}$ and $F_{yf}$ act on the front wheels, while $F_{xr}$ and $F_{yr}$ act on the rear wheels. These forces result from the interaction between the tires and the road surface. A detailed explanation will be provided in the following chapter.

## 3.2 Tire to road surface modeling

To generate tire forces, multitude of models with different complexity can be used. To start, let's define the tire coordinate system. As shown in figure 3.2, the tire coordinate system is a Cartesian system with the x-axis pointing forward, the y-axis to the left, and the z-axis upward. $\dot{\rho}$ is the rotation speed

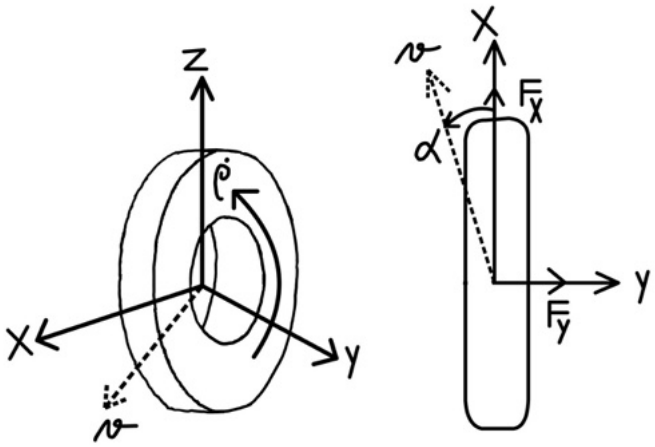



**Figure 3.2:** Tire Coordinate System

of the tire. $\tau$ and $\tau_{Br}$ are the accelerating and braking torques. $\alpha$ is the side-slip angle of the wheel.

### 3.2.1 Wheel dynamics model

We can model the wheel itself with following differential equation:

$$\ddot{\rho} = \frac{1}{J_w}\Big(\tau - \tau_{Br} - F_x \cdot r_w\Big), \tag{3.6}$$

where $J_w$ is the moment of inertia of the wheel, $r_w$ is the radius of the wheel and $F_x$ is the longitudinal force acting on the wheel.

### 3.2.2 Slip variables

Modeling slip of the tire is necessary to capture the drifting behavior of the car. The logitudinal slip ratio $\lambda$ is defined as:

$$\lambda = \frac{\dot{\rho} \cdot r_w - v_x}{\max\{\dot{\rho} \cdot r_w, v_x\}}, \tag{3.7}$$

where $v_x$ is the longitudinal velocity of the wheel. Here we can see that if the wheel will be spinning on spot the slip ratio will be 1. If the wheel is not spinning at all but moving forwards or backwards the slip ratio will be -1.

The lateral slip angle for front wheel is defined as:

$$\alpha_f = -\arctan\left(\frac{(v\sin\beta + l_f\dot{\psi})\cos\delta_f - v\cos\beta\sin\delta_f}{|(v\sin\beta + l_f\dot{\psi})\sin\delta_f + v\cos\beta\cos\delta_f|}\right) \tag{3.8}$$

and for the four-wheel is defined as:

$$\alpha_r = -\arctan\left(\frac{(v\sin\beta - l_r\dot{\psi})\cos\delta_r - v\cos\beta\sin\delta_r}{|(v\sin\beta - l_r\dot{\psi})\sin\delta_r + v\cos\beta\cos\delta_r|}\right) \tag{3.9}$$

### 3.2.3 Bi-linear model

The simplest model is bi-linear model. This model describes the tire force as a function of a slip variable. The model is based on the assumption that the tire forces are linear functions of the slip ratio and slip angle. The model is given by the following equations:

$$F_i = \begin{cases} C \cdot \alpha, & \text{if } |\alpha| < \dfrac{\mu_{\max}}{C} \\ \mu_{\max}, & \text{if } |\alpha| \geq \dfrac{\mu_{\max}}{C} \end{cases}, \tag{3.10}$$

where $F_i$ is the tire force, $C$ is the cornering stiffness, $\alpha$ is the slip angle or slip ratio and $\mu_{\max}$ is the maximum friction coefficient.

### 3.2.4 Pacejka magic formula

The most famous and widely spread tire-to-road surface model is the Pacejka magic formula [23]. For many its the perfect combination of theoretical and

empirical approach. The full model with over 90 parameters was simplified to just 4. The Pacejka magic formula is given by the following equations:

$$F_i = D \cdot F_z \sin\left(C \cdot \arctan\left(B \cdot \alpha - E \cdot (B \cdot \alpha - \arctan(B \cdot \alpha))\right)\right) \tag{3.11}$$

This can be further simplified by setting $E = 0$ to.

$$F_i = D \cdot F_z \sin\left(C \cdot \arctan\left(B \cdot \alpha\right)\right), \tag{3.12}$$

where once again $\alpha$ is the slip angle, $F_z$ is the vertical load on the tire, $D$ is the peak force coefficient, $C$ is the shape factor and $B$ is the stiffness factor.

### 3.2.5 Tire forces interaction

It is necessary to simulate the interaction between longitudinal and lateral forces, as the tire cannot generate maximal force in both directions at the same time. Those interaction are described in chapter 13 of [25] and also in [7]. It is this interconnection between tire forces, which allows power sliding and drifting. Several models are available:

**Empirical approach** This model uses function similar to the magic formula to simulate the interplay of forces. It is used in Liniger's thesis [17]. It depends on the weighing functions of $G_x$ and $G_y$

$$G_x = D_{g,x} \cos\left(C_{g,x} \arctan(B_{g,x}\alpha)\right) \tag{3.13}$$

$$G_y = D_{g,y} \cos\left(C_{g,y} \arctan(B_{g,y}\kappa)\right) \tag{3.14}$$

Notice that $G_x$ is a fuction of slip angle $\alpha$ and $G_y$ is a function of slip ratio $\lambda$. Next we use the weights $G_x$ and $G_y$ to calculate the forces:

$$F_x = G_x \cdot F_{x0} \tag{3.15}$$

$$F_y = G_y \cdot F_{y0} \tag{3.16}$$

,where $F_{x0}$ and $F_{y0}$ are the forces calculated using the magic formula. This aproach is simple to implement and allows more detailed modeling of the interaction between the two forces. However, there are mutiple parameters that need to be calibrated (adjusted?) to get the desired results.

**Traction elipse** this model is assumes that the combined force moves along the elipse. The combined force $F$ that can be generated by tire is given by the following equation:

$$F = \sqrt{\frac{F_x^2}{D_x^2} + \frac{F_y^2}{D_y^2}} \leq \mu \cdot F_z \tag{3.17}$$

,where $\mu$ is the friction coefficient, $F_z$ is the vertical load on the tire. This approach uses the concept of a scaling angle $\beta$, which defines the direction of the total slip vector composed of slip ratio $\lambda$ and slip angle $\alpha$. The angle $\beta$ is computed as:

$$\beta = \arccos\left(\frac{|\lambda|}{\sqrt{\lambda^2 + \sin^2(\alpha)}}\right) \tag{3.18}$$

The actual longitudinal and lateral friction coefficients are calculated from the maximum tire forces and the vertical load:

$$\mu_{x,\text{act}} = \frac{F_{x,\max}}{F_z}, \quad \mu_{y,\text{act}} = \frac{F_{y,\max}}{F_z} \tag{3.19}$$

Maximum attainable friction coefficients (under pure slip) are given by:

$$\mu_{x,\max} = D_x, \quad \mu_{y,\max} = D_y \tag{3.20}$$

The effective friction coefficient in the longitudinal direction is scaled as:

$$\mu_x = \frac{1}{\sqrt{\left(\frac{1}{\mu_{x,\text{act}}}\right)^2 + \left(\frac{\tan(\beta)}{\mu_{y,\max}}\right)^2}} \tag{3.21}$$

The longitudinal force is then scaled accordingly:

$$F_x = \left|\frac{\mu_x}{\mu_{x,\text{act}}}\right| F_{x,\max} \tag{3.22}$$

Similarly, the effective lateral friction coefficient is computed using:

$$\mu_y = \frac{\tan(\beta)}{\sqrt{\left(\frac{1}{\mu_{x,\max}}\right)^2 + \left(\frac{\tan(\beta)}{\mu_{y,\text{act}}}\right)^2}} \tag{3.23}$$

and the lateral force is scaled as:

$$F_y = \left|\frac{\mu_y}{\mu_{y,\text{act}}}\right| F_{y,\max} \tag{3.24}$$

Figure 3.3 shows the combined forces acting on the tire.

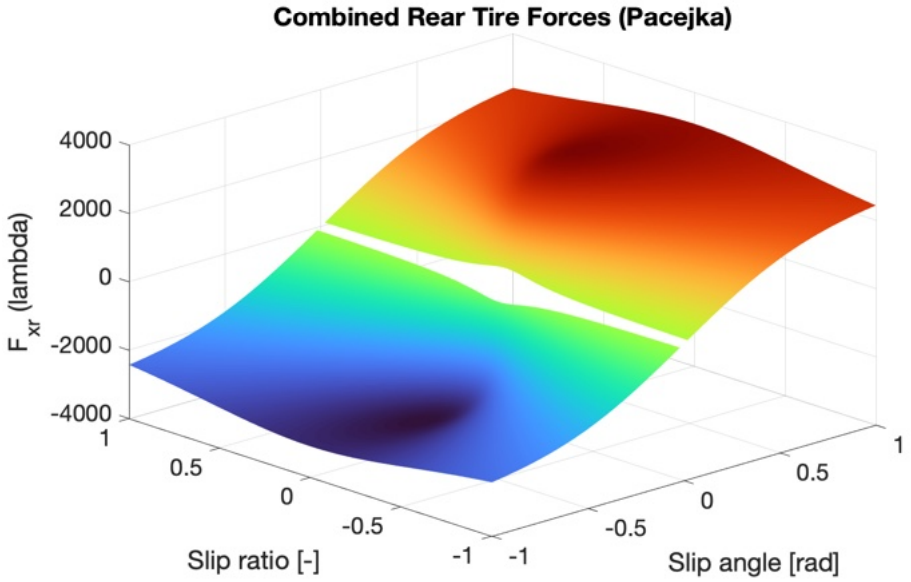



**Figure 3.3:** Combined forces acting on the tire.

# Chapter 4

# Linear Quadratic Control

In this chapter, we focus on the classical *Linear Quadratic (LQ)* optimal control problem. We assume a continuous-time linear time-invariant (LTI) system described by the following state-space equation:

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{4.1}$$

where $x(t) \in \mathbb{R}^n$ is the state vector, $u(t) \in \mathbb{R}^m$ is the control input, and $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$ are system matrices.

The goal is to design a state-feedback control law $u(t) = -Kx(t)$ that minimizes the infinite-horizon quadratic cost function:

$$J = \int_0^\infty \left( x(t)^\top Q x(t) + u(t)^\top R u(t) \right) dt, \tag{4.2}$$

where $Q \in \mathbb{R}^{n \times n}$ and $R \in \mathbb{R}^{m \times m}$ are weighing matrices. We assume that:

- $Q$ is symmetric and positive semi-definite: $Q = Q^\top \succeq 0$,
- $R$ is symmetric and positive definite: $R = R^\top \succ 0$.

These matrices define the trade-off between minimising the state error and penalizing control effort. The optimal state-feedback control law that minimizes the cost function is given by:

$$u(t) = -Kx(t), \tag{4.3}$$

where the gain matrix $K$ is computed as:

$$K = R^{-1}B^{\top}P, \tag{4.4}$$

and $P \in \mathbb{R}^{n\times n}$ is the unique positive semi-definite solution to the Continuous-time Algebraic Riccati Equation (CARE):

$$A^{\top}P + PA - PBR^{-1}B^{\top}P + Q = 0. \tag{4.5}$$

Solving the CARE yields the matrix $P$, which defines the optimal trade-off between state performance and control effort. The resulting feedback gain $K$ guarantees that the closed-loop system $\dot{x}(t) = (A - BK)x(t)$ is asymptotically stable, provided that the system is controllable and the cost matrices $Q$ and $R$ satisfy the assumptions stated above.

## 4.1 Extension to Reference Tracking

While the standard LQR formulation is designed for regulation—bringing the system state to the origin—it can be extended to perform reference tracking, where the goal is to follow a desired time-varying or constant reference trajectory.

Let the reference state be denoted by $x_{\text{ref}}(t)$. The tracking problem then aims to minimize the deviation from this reference:

$$e(t) = x(t) - x_{\text{ref}}(t). \tag{4.6}$$

Assuming that the reference trajectory $x_{\text{ref}}(t)$ is generated by a known nominal input $u_{\text{ref}}(t)$, we define the error dynamics as:

$$\dot{e}(t) = Ae(t) + Bu(t). \tag{4.7}$$

This formulation is structurally equivalent to the regulation problem, and thus the same LQR gain $K$ can be applied to the error state:

$$u(t) = K(x(t) - x_{\text{ref}}(t)). \tag{4.8}$$

This control law stabilizes the error dynamics around zero, effectively tracking the reference trajectory.

## 4.2 Nonlinear singletrack model

The nonlinear sigle track described erlier was used and adopted to design LQR regulator. State vector was carried over and used in following form:

$$\mathbf{x} = \begin{pmatrix} v_x \\ v_y \\ \dot{\psi} \end{pmatrix}. \tag{4.9}$$

To simplify the design proces following changes were done. Instead of usign the input vector as decribed erlier, the control variable was changed to wheel speed $\dot{\rho}$. This simplification still alows the modeling of wheel slip but simplfies the wheel dynamics. With this change any braking torque and acceleration torque is substituted by the wheel speed. The $F_x$ force of the front tire was neglected. This simplification is made possible as only rear wheel driven car will be assumed so the front force $F_x$ would be close to zero. This will however afect the braking ability of the car as it virtauly has only rear breaks. The input vector is now:

$$\mathbf{u} = \begin{pmatrix} \delta_f \\ \dot{\rho}_r \end{pmatrix} \tag{4.10}$$

Applying the mantioned simplifications we can write the following equations of motion:

$$\begin{aligned} \dot{v}_x &= \frac{1}{m}\Big(F_{xr} - F_{yf}\sin\delta_f\Big) - \dot{\psi}\, v_y, \\ \dot{v}_y &= \frac{1}{m}\Big(F_{yf}\cos\delta_f + F_{yr}\Big) + \dot{\psi}\, v_x, \\ \ddot{\psi} &= \frac{1}{I_z}\Big(l_f F_{yf}\cos\delta_f - l_r F_{yr}\Big) \end{aligned} \tag{4.11}$$

To simulate tire forces I used the Pacejka magic formula as described in 3.11, when also incorporating the traction elipse model for rear tire. I also approximate side-slip angles using folowing equations:

$$\begin{aligned} \alpha_f &\approx \delta_f - \arctan\left(\frac{\dot{\psi}\, l_f + v_y}{v_x}\right) \\ \alpha_r &\approx \arctan\left(\frac{\dot{\psi}\, l_r - v_y}{v_x}\right) \end{aligned} \tag{4.12}$$

### 4.2.1 Model parameters

The model parameters were set for a middle class passanger car on a slightly slippery road. Model parametrs are presented in the following table 4.1.

| Parameter | Value | Unit |
|---|---|---|
| Mass ($m$) | 1200 | kg |
| Moment of Inertia ($I_z$) | 1388 | kg $\cdot$ m$^2$ |
| Distance from CG to front ($l_f$) | 1.4 | m |
| Distance from CG to rear ($l_r$) | 1.6 | m |
| Wheel radius ($r_w$) | 0.33 | m |
| **Front tire Pacejka coeficients** | | |
| $C$ | 4.15 | [-] |
| $B$ | 2.00 | [-] |
| $D$ | 0.72 | [-] |
| **Rear tire Pacejka coeficients** | | |
| $C_y$ | 2.00 | [-] |
| $B_y$ | 4.15 | [-] |
| $D_y$ | 0.60 | [-] |
| | | |
| $C_x$ | 1.40 | [-] |
| $B_x$ | 5.00 | [-] |
| $D_x$ | 0.60 | [-] |

**Table 4.1:** Model Parameters

## 4.3 Model equilibrium

To design LQR a linearized model is needed. The model will be linearized around its equilibrium points as this provides minimized loss of generality. At the system's equilibrium, all the state derivatives are zero. We can now write the model equations in the following form:

$$\begin{aligned} 0 &= \frac{1}{m}\left(F_{xr}^{eq} - F_{yf}^{eq}\sin\delta_f^{eq}\right) - \dot{\psi}^{eq}\, v_y^{eq}, \\ 0 &= \frac{1}{m}\left(F_{yf}^{eq}\cos\delta_f^{eq} + F_{yr}^{eq}\right) + \dot{\psi}^{eq}\, v_x^{eq}, \\ 0 &= \frac{1}{I_z}\left(l_f F_{yf}^{eq}\cos\delta_f^{eq} - l_r F_{yr}^{eq}\right) \end{aligned} \tag{4.13}$$

We have obtained set of three equations with equlibrium values. We can use the equations 3.11 to investigate further. Equilibrium values of the forces are given by:

$$\begin{aligned} F_{yf}^{eq} &= D \cdot F_z \sin\left(C \cdot \arctan\left(B \cdot \alpha_f^{eq}\right)\right) \\ F_{xr}^{eq} &= D \cdot F_z \sin\left(C \cdot \arctan\left(B \cdot \lambda_r^{eq}\right)\right) \\ F_{yr}^{eq} &= D \cdot F_z \sin\left(C \cdot \arctan\left(B \cdot \alpha_r^{eq}\right)\right) \end{aligned} \tag{4.14}$$

Further rewriting the equations using the equations 4.12 and equation 3.7,

we can write the equilibrium values as:

$$\begin{aligned}
\alpha_f^{eq} &= \delta_f^{eq} - \arctan\left(\frac{\dot{\psi}^{eq}\, l_f + v_y^{eq}}{v_x^{eq}}\right) \\
\alpha_r^{eq} &= \arctan\left(\frac{\dot{\psi}^{eq}\, l_r - v_y^{eq}}{v_x^{eq}}\right) \\
\lambda_r^{eq} &= \frac{\dot{\rho}_r^{eq} \cdot r_w - v_x^{eq}}{\max\{\dot{\rho}_r^{eq} \cdot r_w, v_x^{eq}\}}
\end{aligned} \tag{4.15}$$

We have now defined five equlibrium values $v_x^{eq}$, $v_y^{eq}$, $\dot{\psi}^{eq}$, $\delta_f^{eq}$, $\dot{\rho}_r^{eq}$. To find the equilibrium values we need to solve the system of equations. The values of $v_x^{eq}$ and $\delta_f^{eq}$ were fixed and then the system of equations was solved. Matlab and the fsolve function was used. Finding rasonable solutions proved to be dificult as the function fsolve expects initial guess of solution. As the model is nonlinear, the initial guess was very important. The value of $v_x^{eq}$ was set to 8.0 m/s. The value of $\delta_f^{eq}$ was set to an interval from $-25$ to 25 degrees. The values of $v_y^{eq}$, $\dot{\psi}^{eq}$ and $\dot{\rho}_r^{eq}$ were calculated using the fsolve function. Its output is in figure 4.1.

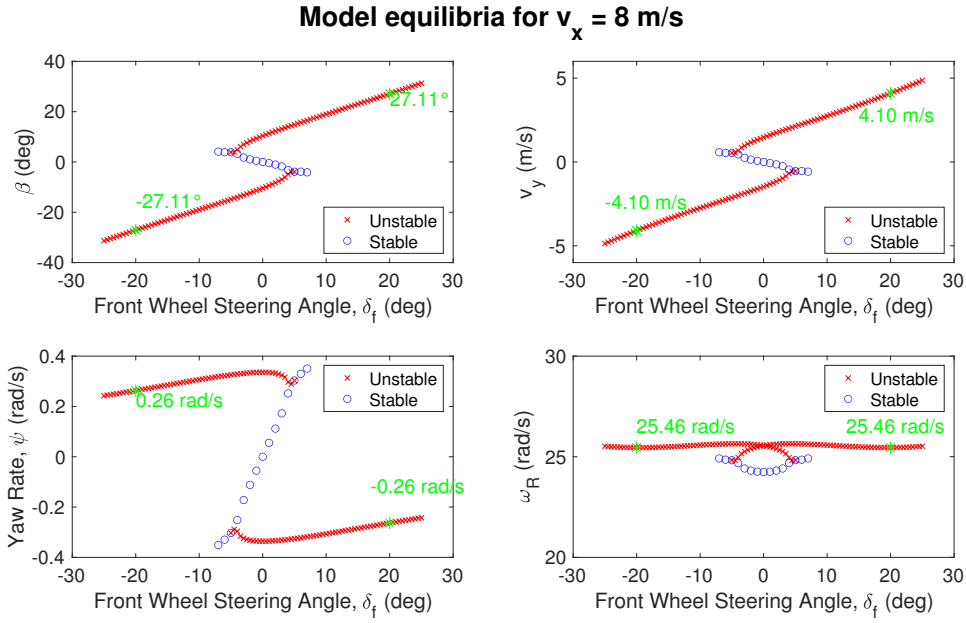



**Figure 4.1:** Equilibrium points of the vehicle dynamics model

We can see stable and unstable equilibrium points. The stable region of normal driving is characterized by not having an excesive slip as displayed in blue in the figure. than we have the unstable reagions of understeer and oversteer. The region of oversteer is where I want to linearize my model.

## 4.4 Linearization

In control design, particularly for methods like Linear Quadratic Regulator (LQR), it is necessary to work with a linearized model of the system around an operating point (equilibrium). The linearized system dynamics are given by:

$$\dot{x} = Ax + Bu \tag{4.16}$$

where the matrices $A$ and $B$ are the Jacobians of the system dynamics $f(x, u)$ with respect to the state $x$ and the input $u$ respectively:

$$A = \left.\frac{\partial f}{\partial x}\right|_{x_0,u_0}, \quad B = \left.\frac{\partial f}{\partial u}\right|_{x_0,u_0} \tag{4.17}$$

Folowing equilibrium point was chosen as an operating point:

$$\begin{aligned} v_x^{eq} &= 8.00\frac{m}{s} \\ v_y^{eq} &= -4.09\frac{m}{s} \\ \dot{\psi}^{eq} &= 0.26\frac{rad}{s} \\ \delta_f^{eq} &= -20^\circ \\ \dot{\rho}_r^{eq} &= 25.46\frac{rad}{s} \approx 243\text{RPM} \end{aligned} \tag{4.18}$$

### 4.4.1 Numerical Linearization via Finite Differences

When an analytical derivation of $A$ and $B$ is complicated or impractical, a numerical method known as **finite difference approximation** can be used. Main advantage of this method is its simplicity and flexibility, especially for complex models. However, it is important to note that this method is only an approximation and may introduce numerical errors. Getting analytical solution is always more precise. This method estimates the partial derivatives by perturbing each state and input variable individually by a small amount $\epsilon$. The approximation for the $i$-th column of $A$ is:

$$A_{:,i} \approx \frac{f(x_0 + \epsilon e_i, u_0) - f(x_0, u_0)}{\epsilon} \tag{4.19}$$

where $e_i$ is the $i$-th **canonical basis vector**, defined as:

$$e_i = [0, \ldots, 1, \ldots, 0]^T \tag{4.20}$$

with a 1 in the $i$-th position and zeros elsewhere. Similarly, the approximation for the $i$-th column of $B$ is:

$$B_{:,i} \approx \frac{f(x_0, u_0 + \epsilon e_i) - f(x_0, u_0)}{\epsilon} \tag{4.21}$$

This method computes the partial derivatives by applying a small perturbation to only one variable at a time, leaving all others unchanged. To perform finite differencing correctly, it is important to perturb along the canonical directions in the state and input spaces. In this case the state vector is:

$$x = [v_x, v_y, \dot{\psi}]^T \tag{4.22}$$

then the canonical basis vectors are:

$$e_1 = [1, 0, 0]^T \tag{4.23}$$
$$e_2 = [0, 1, 0]^T \tag{4.24}$$
$$e_3 = [0, 0, 1]^T \tag{4.25}$$

Perturbing with $\epsilon e_1$ means increasing only $v_x$ by $\epsilon$, while keeping $v_y$ and $\dot{\psi}$ unchanged, and so forth. For this model, a perturbation of size $\epsilon = 1 \times 10^{-6}$ was used. The control input vector wasl limited to only the steering angle.

$$u = [\delta_f] \tag{4.26}$$

The resulting Jacobian matrices $A$ and $B$ are:

$$A = \begin{bmatrix} -0.9604 & -0.4493 & -5.2017 \\ -0.0923 & 0.2456 & -8.3930 \\ -1.4663 & -2.9900 & -3.1667 \end{bmatrix} \qquad B = \begin{bmatrix} 5.0331 \\ 0 \\ 25.8008 \end{bmatrix}$$

Verification simulation results are in figure 4.2. As we can see both systems behave similarly. The resulting linearized system is controlable as the controllability matrix has full rank. this means that the main codnition for LQR design is satisfied.

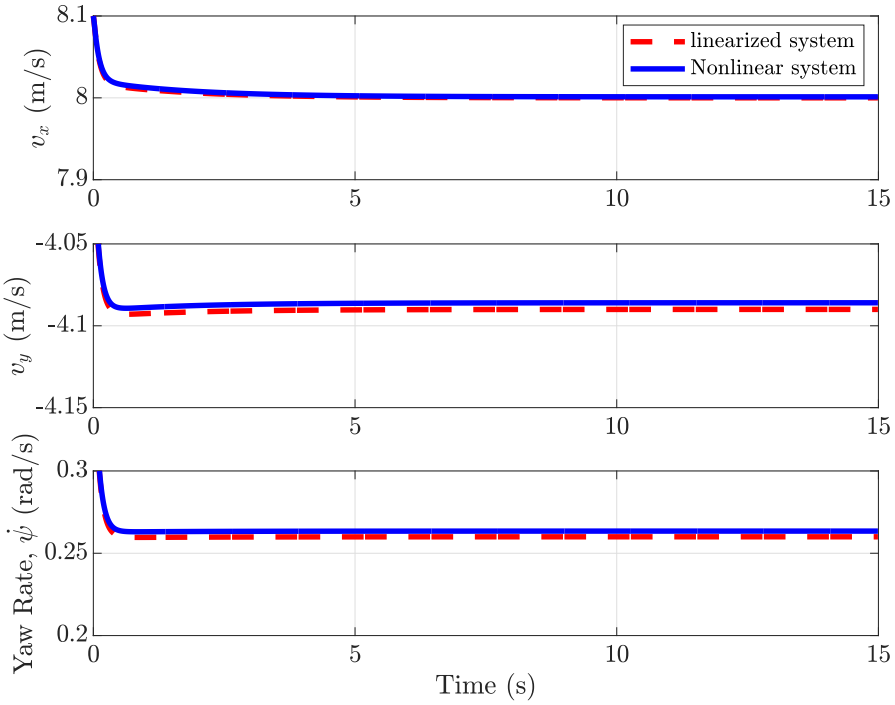



**Figure 4.2:** Comparison of the nonlinear and linearized system responses.

# Chapter 5

# Controller design for classical platform

## 5.1 Control architecture

The LQR with reference tracking is used. It controls only the steering angle. The wheel speed is controlled by PI controller. In total, three LQR gains are generated. One for the normal driving region and two for drifting regions. PI controller values are the same for all three regions. Referance state vector will be generated by path tracking algorithm. An overview of the control architecture is shown in figure 5.1.

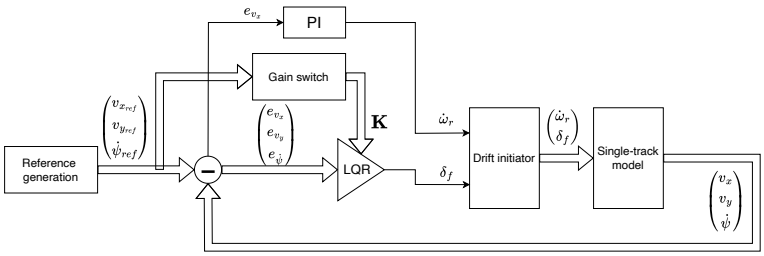



**Figure 5.1:** Block diagram of the 2WS control structure.

## 5.2 PI controller design

PI controller was chosen to control longitudinal velocity of the car. For the design of the controller, the linearised model in the normal driving region was used with following equilibrium point:

$$\begin{aligned} v_x^{eq} &= 8.00\frac{m}{s} \\ v_y^{eq} &= 0.00\frac{m}{s} \\ \dot{\psi}^{eq} &= 0.00\frac{rad}{s} \\ \delta_f^{eq} &= 0.00rad \\ \dot{\rho}_r^{eq} &= 24.24\frac{rad}{s} \approx 230\text{RPM} \end{aligned} \tag{5.1}$$

Resulting $A$ and $B$ matrices from linearization are:

$$A = \begin{bmatrix} -2.4032 & 0 & 0 \\ 0 & -2.8498 & -3.4403 \\ 0 & -0.7884 & -12.9301 \end{bmatrix} \qquad B = \begin{bmatrix} 0.7931 & 0 \\ 0 & 0 \\ 0 & 37.8441 \end{bmatrix}$$

The assumed control input vector is as follows:

$$u = [\delta_f \quad \dot{\rho}_r] \tag{5.2}$$

For the PI controller only SISO system was used reducing the input vector to:

$$u = [\dot{\rho}_r] \tag{5.3}$$

The controlled variable is longitudinal speed $v_x$. Its output is the wheel speed $\dot{\rho}_r$. The structure is shown in figure 5.2.

The PID design was done using root locus tool in Matlab. The open loop transfer functio for PI controller is:

$$C(s) = K_p + \frac{K_i}{s} = \frac{K_p s + K_i}{s} \tag{5.4}$$

Linearized model transfer function is:

$$G(s) = \frac{0.7931}{s + 2.403} \tag{5.5}$$

In the root locus tool additional pole was placed to zero and additional zero was placed at $-2$. Design requirement were set for over shoot of maximal 5% and settling time of 2 s. The resulting PI controller transfer function is:

$$C(s) = \frac{55 \cdot (s+2)}{s} = 55 + \frac{110}{s} \tag{5.6}$$

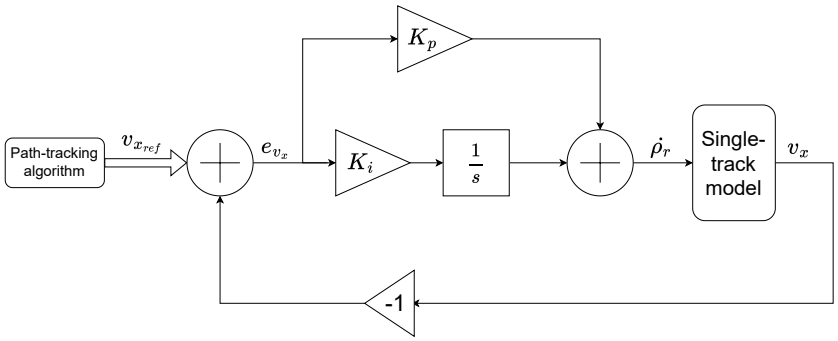



**Figure 5.2:** Block diagram of the PID controller.

PI coefficients are $K_p = 55$ and $K_i = 110$ and the closed loop transfer function is:

$$T(s) = \frac{43.62s + 87.25}{s^2 + 46.03s + 87.25} \tag{5.7}$$

Simulation results for PI controller and nonlinear system are in figure 5.3. There is a overshoot present the antiwindup function was used however the overshoot was not eliminated.

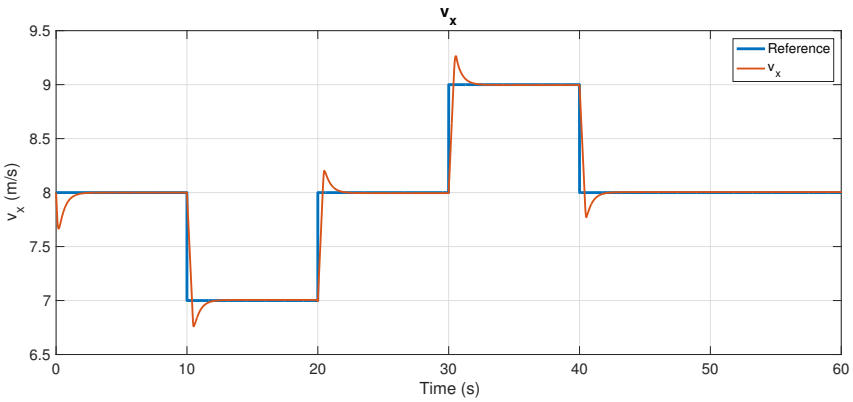



**Figure 5.3:** Simulation results of the nonlinear system with PI controller.

## 5.3 LQR design

Three different operating points have been chosen. The longitudinal velocity is fixed at $v_x = 8.0\frac{m}{s}$. The first two points are chosen in the drifting region, one for drifting with side-slip angle $\approx -30$ degrees the other for side-slip angle $\approx 30$ degrees. The third point is chosen in the normal driving region with zero side-slip angle and yaw rate. The equilibrium points are:

| **Left drift** | **Right drift** | **Straight driving** |
|---|---|---|
| $\delta_{f,\text{eq}} = -0.3491\,\text{rad}$ | $\delta_{f,\text{eq}} = 0.3491\,\text{rad}$ | $\delta_{f,\text{eq}} = 0\,\text{rad}$ |
| $v_{y,\text{eq}} = -4.09\,\text{m/s}$ | $v_{y,\text{eq}} = 4.09\,\text{m/s}$ | $v_{y,\text{eq}} = 0\,\text{m/s}$ |
| $v_{x,\text{eq}} = 8.0\,\text{m/s}$ | $v_{x,\text{eq}} = 8.0\,\text{m/s}$ | $v_{x,\text{eq}} = 8.0\,\text{m/s}$ |
| $r_{\text{eq}} = 0.26\,\text{rad/s}$ | $r_{\text{eq}} = -0.26\,\text{rad/s}$ | $r_{\text{eq}} = 0\,\text{rad/s}$ |
| $\omega_{R,\text{eq}} = 25.46\,\text{rad/s}$ | $\omega_{R,\text{eq}} = 25.46\,\text{rad/s}$ | $\omega_{R,\text{eq}} = 24.24\,\text{rad/s}$ |

Linearization was done using the finite difference method as described in previous chapter. The choice of values for the weighting matrices $Q$ and $R$ is crucial for the performance of the LQR controller. For drifting regions, the $Q$ matrix was constructed to penalize deviations of the system states. This metod was also used in [35]. Each diagonal element of the $Q$ matrix was computed as:

$$Q_{ii} = \left| \frac{x_i^{\text{eq}}}{\Delta x_i^{\max}} \right| \tag{5.8}$$

where:

- $x_i^{\text{eq}}$ is the equilibrium value of the $i$-th state,
- $\Delta x_i^{\max}$ is the maximum acceptable deviation of that state from equilibrium.

This strategy ensures that more critical states, or states with smaller allowable variations, are weighted more heavily in the cost function. It also provides a dimensionally consistent and systematic way to construct the $Q$ matrix.

The resulting $Q$ matrix for the three operating points is:

$$Q = \begin{bmatrix} \left|\frac{v_x^{\text{eq}}}{0.8}\right| & 0 & 0 \\ 0 & \left|\frac{v_y^{\text{eq}}}{0.5}\right| & 0 \\ 0 & 0 & \left|\frac{\dot{\psi}^{\text{eq}}}{0.02}\right| \end{bmatrix} = Q = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 8.18 & 0 \\ 0 & 0 & 13 \end{bmatrix} \tag{5.9}$$

Using equation 5.8 for setting the $Q$ matrix values base on the equilibrium points, works suitably for drifting regions. However, for the normal driving region, the resulting matrix would be zero for elements $Q_{22}$ and $Q_{33}$. To avoid this the values of $Q_{22}$ and $Q_{33}$ set to values of 8 and 13 respectively. The values were chosen based on the drifting regions.

The $R$ matrix was chosen with value of 5 for drifting and normal driving region. Higher value was set for smoother control input. The resulting $R$ matrix is:

$$R = \begin{bmatrix} 5 \end{bmatrix} \tag{5.10}$$

Final computation of the gain matricies was done using the Matlab function lqr. The resulting gain matrices are:

$$K_L = \begin{bmatrix} -0.3909 \\ -1.4474 \\ 1.8868 \end{bmatrix}, \quad K_R = \begin{bmatrix} 0.3909 \\ -1.4474 \\ 1.8869 \end{bmatrix}, \quad K_S = \begin{bmatrix} 0.000 \\ -0.552 \\ 1.3479 \end{bmatrix}$$

## 5.4 Drift initiation

The control architecture has to switch between diffrent K matricies and intiate the drift maneuver. For the drift intiation I was inpired by powersliding. When set level of reference for yaw rate and side-slip angle is reached, the car will apply full steering lock and full throttle. The state diagram is shown in figure 5.4.

The drift is initiated when controller recieves the absolute value of yaw rate reference higher than $0.2 rad/s$ and the slip angle on rear tire $\alpha_r$ is lower than 0.3. While those two conditions are met the output value for wheel speed is set to 60 rad/s. The steering angle is set to $\pm 0.3$ rad depending on the

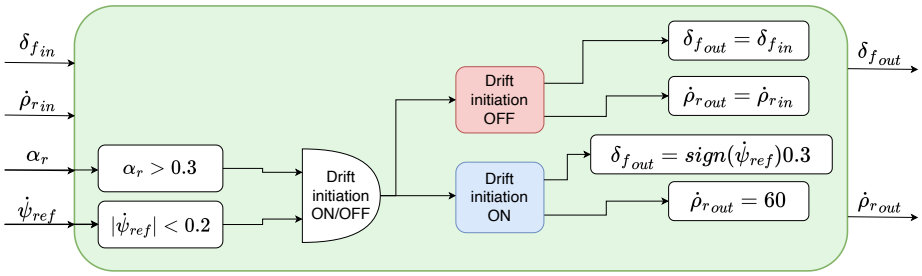



**Figure 5.4:** Drift initiator state diagram.

direction of the drift. When one of the conditions is not met the controller continues to send its input values and does not interrupt the drift.

The value limit for slip angle $\alpha_r$ is set to 0.3 is chosen based on the Pacejka model. As can be seen in figure 5.5 the force generated by the tire saturate around value 0.24. When the $\alpha_r$ reaches the value of 0.3 the system is in somewhat linear region and the linear controller can be switched and used.

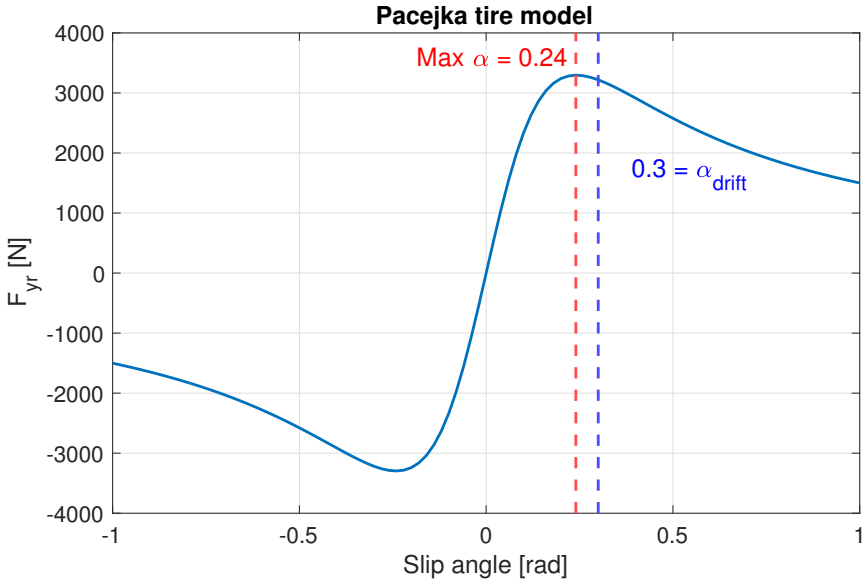



**Figure 5.5:** The limits of the Pacejka model, illustrating the range of slip angles where the tire force saturates.

## 5.5 Simulation results

The reference values for the control structure were generated arbitrarily. Sixty second simulation was run. During simulation the car changes from normal

driving to drifting and back to normal driving. Between 10 and 30 seconds the car is in the drifting region and performing leftside drift. Between 30 and 40 seconds, the car settled to normal driving. For the final period, between 40 and 60 seconds, the car is in the right side drift.

In figure 5.6 we can see the vehicle state vector and input vector during simulation. We can also see its ability to track the reference values. Constant tracking error can be seen for the yaw rate and velocity in y during drifting stages of the simulation. No constant tracking error is present in the velocity in x direction. From the figure, we can clearly see that drift initiation and drift stabilization was done successfully. The car is able to change its sideslip angle while maintaining the desired yaw rate.

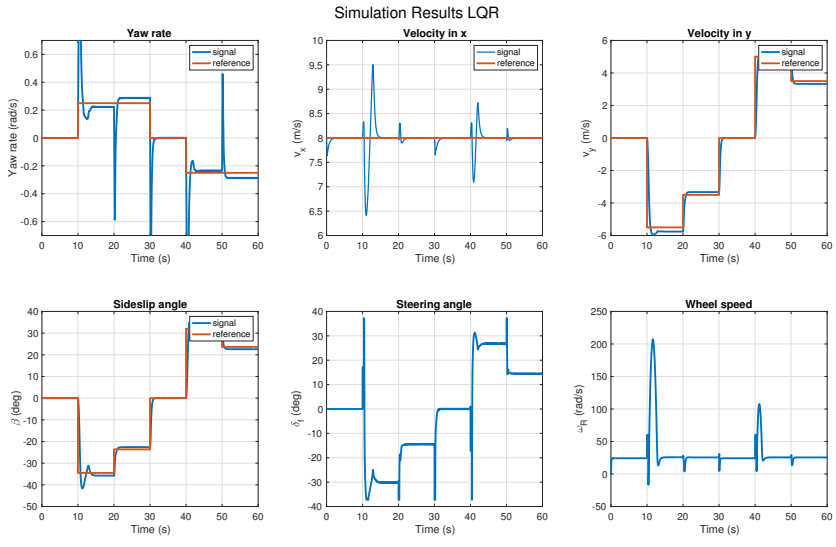



**Figure 5.6:** Vehicle state vector and input vector during simulation.

Figure 5.7 presents slip variables of rear wheel. Here we can see that during the drift initiation and settling the slip ratio is well above safe limits. This indicate saturation of the tire. When settled in the drift maneuver the slip ratio is lowers to the one of normal driving. The slip angle of the tire is well above the saturation level during the drifting maneuver.

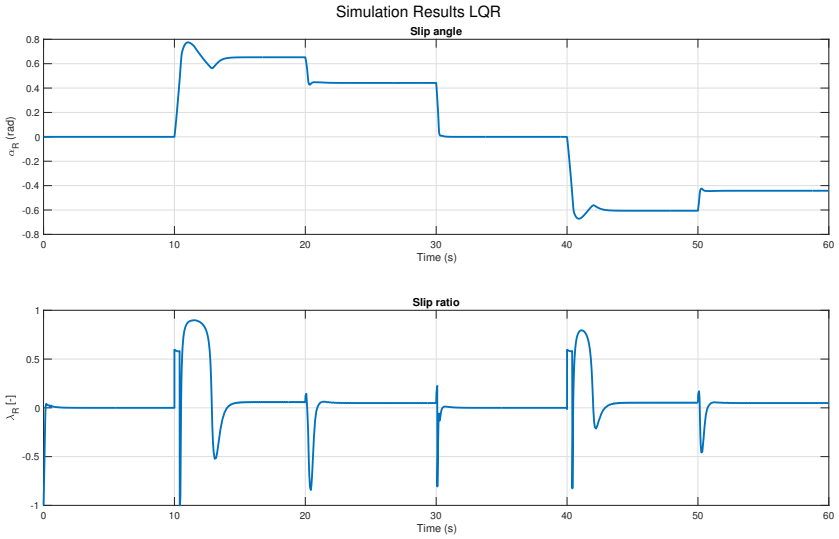



**Figure 5.7:** Rear weel slip variables simulation results.

Figure 5.8 presents the car position in world frame. The drifting sections are higlighted. We can see that the car did not spin and that the drift was initiated and settled multiple times.

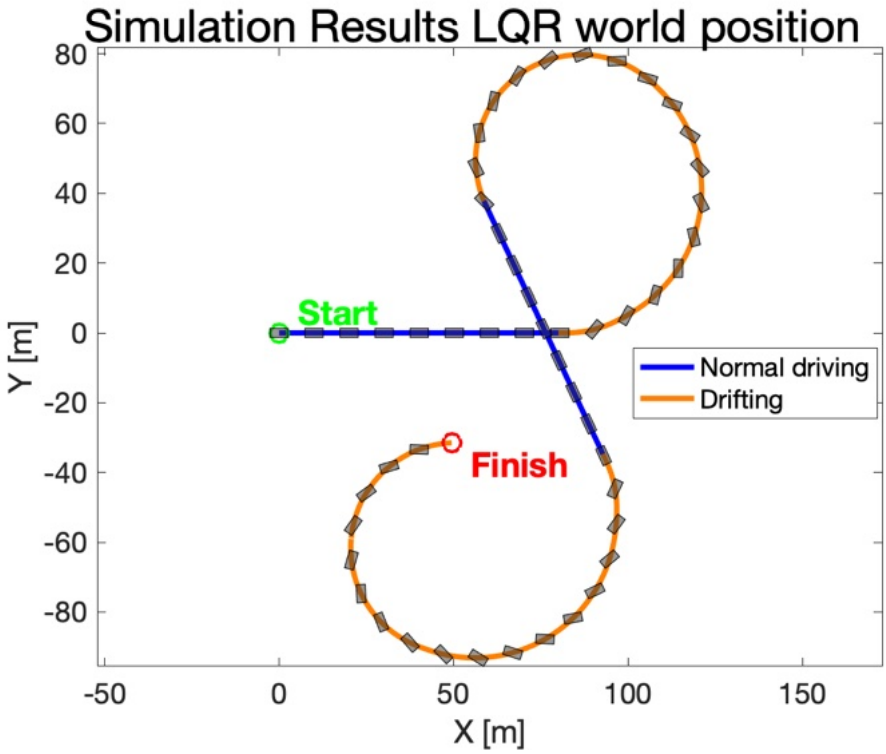



**Figure 5.8:** Car position in world frame during simulation.

Second run simulation was done with ramp change of the $v_y$ reference while maintaining the yaw rate reference. The ramp change was done for both left and right side drift maneuver. In figure 5.9 we can see the car once again tracking the reference with some degree of error. The car reaches maximal side-slip angle around 40 degrees. Countersteering can be seen after the drift initiation. During the ramp response we can see that wheel speed has steadily grown. This suggests an increase in the slip ratio, thus moving further on the slip curve and limiting the generated longitudinal force on the rear tire. This enables the rear tire to slip more in the lateral direction.

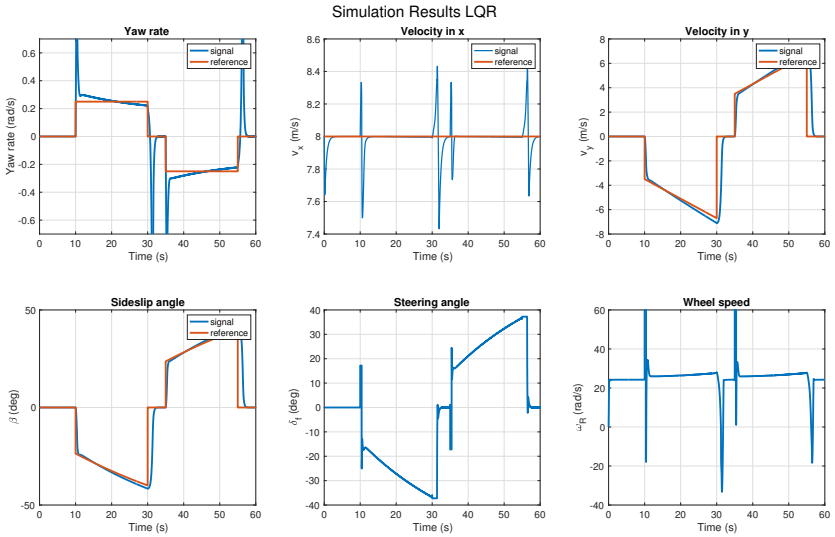



**Figure 5.9:** Vehicle state vector and input vector during simulation.

Figure 5.10 presents slip variables of rear wheel. Here we can see that during the drift initiation, the wheel is once again oversaturated. Both slip angle and slip ratio settled during the change from drifting to normal driving. We can see drop of slip ratio to values close to minus one. This indicates full wheel-lockup.

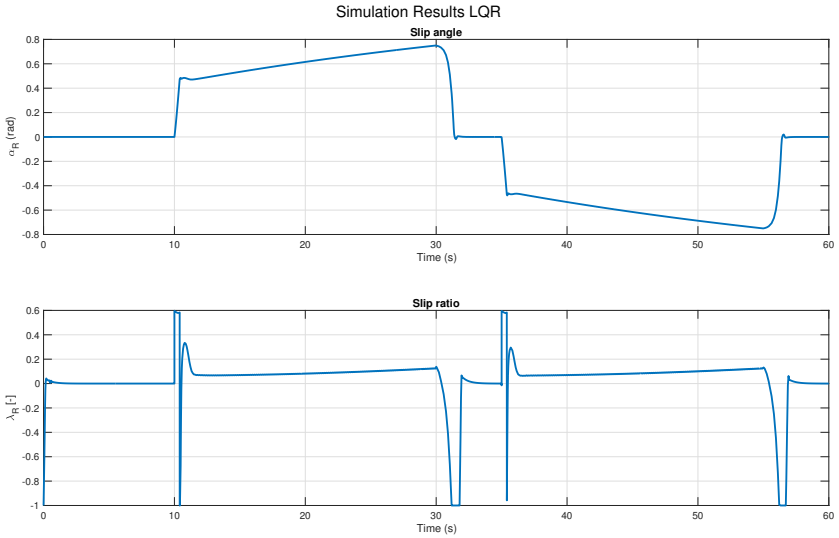



**Figure 5.10:** Rear weel slip variables simulation results.

The vehicle position in world frame is plotted in figure 5.11. The car enters the drift maneuver and does not spinout. Its also able to settle back to normal driving while switching between the LQR gain matrices.

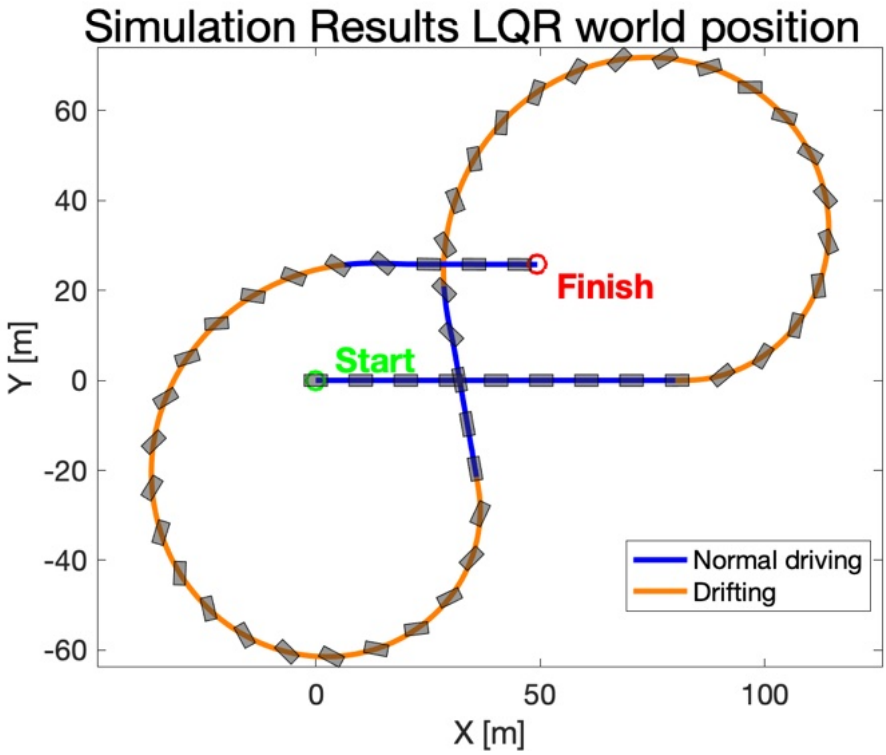



**Figure 5.11:** Car position in world frame during simulation.

## 5.6 Visualization

To enhance the interpretability of the developed control algorithms and to provide an intuitive understanding of the vehicle behavior during various maneuvers, a custom visualization environment was implemented using the Godot Engine [5]. Godot is an open-source, lightweight game engine that supports real-time 3D rendering, scripting, and animation, making it suitable for visualizing dynamic simulation data.

The visualization pipeline begins with the export of simulation results from MATLAB in the form of CSV files containing time-series data of vehicle states such as position, velocity, orientation, and steering angles. Selected simulation file is than loaded by a python script that converts the data into a format compatible with Godot. The data are streamed to the Godot engine using TCP sockets, allowing for real-time visualization of the vehicle. The option to stream real-time data during a simulation was considered but was not implemented due to time constraints.

Different tracks can be created with different textures as presnted in

figure 5.12.

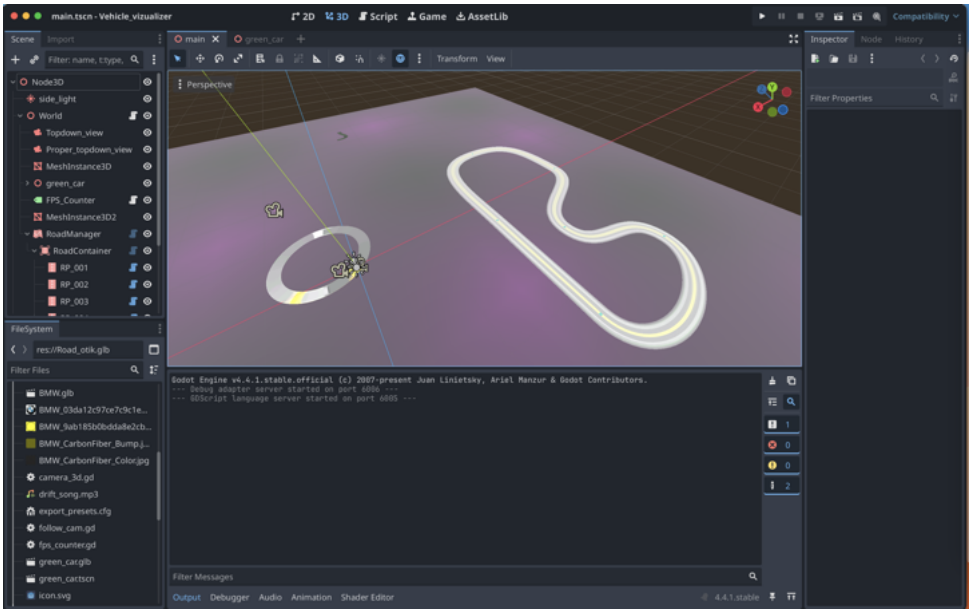



**Figure 5.12:** Godot Engine.

The full simulation can be accessed on YouTube at `https://youtu.be/e11fYA8xFN8`. Here a frame is presented in figure 5.13. We can see the 2WS platform mid drift countersteering.

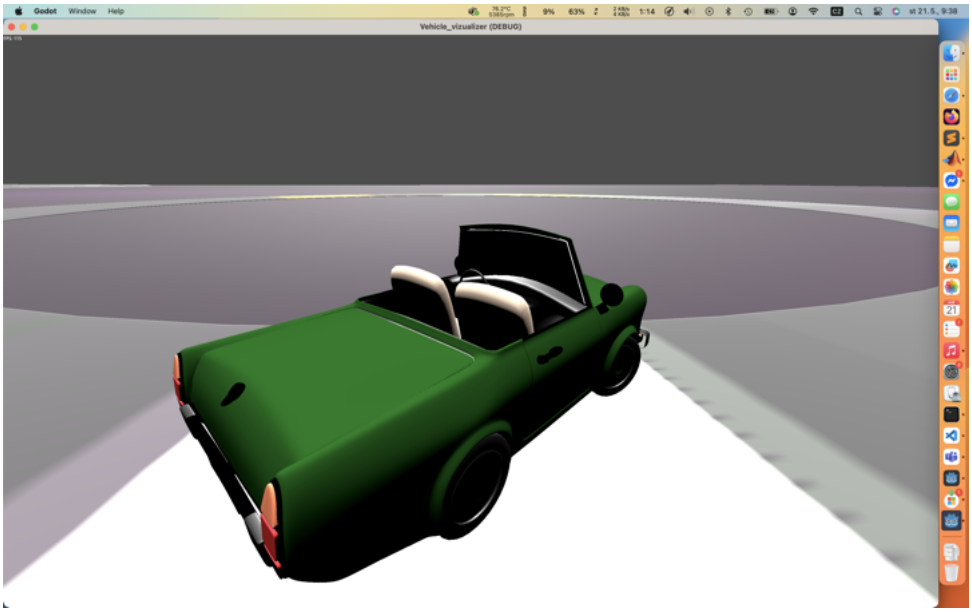



**Figure 5.13:** Godot Engine.

# Chapter 6

# Four-wheel overactuated platform

By gaining another degree of freedom, the variety of control strategies has increased. From state of the art chapter, we can see multiple approaches to a control of four-wheel drive platform. Those range from PID family, continuing to LQR and MPC and finishing with reinforcement learning controllers. The control architecture chosen for the four-wheel platform is shown in figure 6.1. Architecture uses PID controllers to control decoupled lateral and longitudinal dynamics. Here a control allocation problem can be well described. A control allocation allows for modularization of the control system. Higher level controller can be used to specify car's yaw rate, acceleration or velocity and the low-level control scheme distributes this comanding action to individual wheels by commanding appropriate wheel's slip variables. An example can be a yaw stabilization where the desired yaw angle can be obtained from optimisation. The low-level controller than distributes the desired yaw to individual wheels. The low-level controller can be a PID or a LQR controller. At CTU, the problem was studied hevily by Ing. David Vošáhlík, Ph.D. in his work [33] on longitudinal dynamics

## 6.1 Model changes

As I received the control architecture from SDS team at CTU, the model used was a single track model. In comparison to the 2WS platfrom, the single track used for 4WS simulation has modeled the wheel dynamics as described in 3.6. Furthermore, the front longitudinal force is not neglected. The state vector of the model is also changed to the $v, \beta, \dot{\psi}$ notation. The calculation

to the previously used notation was implemented. As the wheel dynamics are modeled, the input vector has changed to:

$$\mathbf{u} = \begin{pmatrix} \delta_f \\ \delta_r \\ \tau_f \\ \tau_r \end{pmatrix} \tag{6.1}$$

The model has the same implementation of the Pacejka tire model as well as the traction elipse model.

## 6.2 Control architecture

The control architecture is shown in figure 6.1. The reference value of velocity vector and yaw rate is transformed to reference values of acceleration and yaw acceleration. These references are changed by transformation function to the reference values of acceleration for each wheel. The longitudinal part, as presented by [33], is controled by cascade of PID controllers. The first PID controller has as a reference desired acceleration in x direction. Its output is used as a slip-ratio reference for the second PID controller. Its output is the wheel speed torque for a specific wheel. The lateral dynamics are controled using the same approach with two PIDs.

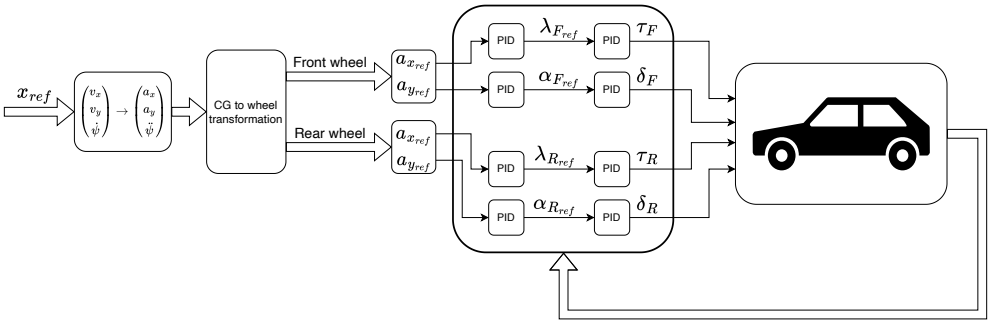



**Figure 6.1:** Control structure architecture of the overactuated platform.

## 6.3 PI values tuning

### 6.3.1 Longitudinal dynamics

As mentioned in the previous chapter, the PI controllers were used to control longitudinala and lateral dynamics. The tuning process was started with the longitudinal dynamics. Initial state vas set to $v_x = 4.0$ m/s with reference set to $v_x = 10.0$ m/s. Using step responces the transfer functions were identified. Tuning was started with the inner loop oflongitudinal dynamics. Here slip ratio reference is tansfered to wheel speed reference. The output of the PI controller is the torque on wheel. The transfer function from torque on wheel to the slip ratio was identified. The transfer function is:

$$G_\lambda(s) = \frac{1.7 \times 10^{-4}}{0.05s + 1} \tag{6.2}$$

Using root locus tool in Matlab the PI controller was tuned. Values of the PI controller were set to $K_p = 8 \cdot 10^3$ and $K_i = 9 \cdot 10^4$. The close loop transfer function is:

$$G_{\lambda\text{closed}}(s) = \frac{1.36s + 15.3}{0.05s^2 + 2.36s + 15.3} \tag{6.3}$$

Poles are located at $-39.44$ and $-7.75$. The system is stable with 0% overshoot and settling time of 0.38 s.

To tune the outer loop the transfer function from slip ratio to the longitudinal acceleration was identified. The trasfer fuction was identified from intial rise of the acceleration only. As resistence is moddled based on the speed the acceleration decreases with increasing speed however the time constant is much higer so thie efect is ignored. For tuning, the system is modeled as second order and its transfer function is:

$$G_{a_x}(s) = \frac{9.956 \times 10^4}{s^2 + 266.7\,s + 7.111 \times 10^4} \tag{6.4}$$

For the outer loop the values of the PI controller were set to $K_p = 0.1$ and $K_i = 2$. The closed-loop system exhibits a pair of complex conjugate dominant poles located at $-132.1 \pm j250.9$ and a third real pole at $-2.48$, indicating a fast underdamped second-order response with an additional slower stable mode. In figure 6.2 we can see the step response of the closed loop system.

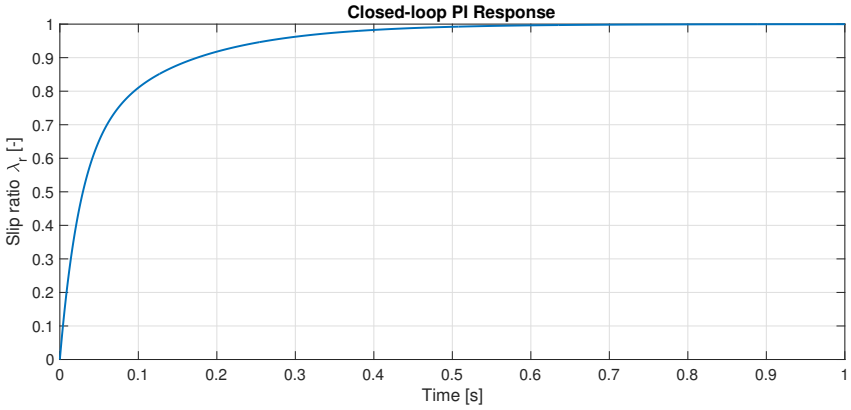



**Figure 6.2:** Step response of close loop.

### 6.3.2 Lateral dynamics

In lateral dynamics loop, the inner loop has a slip angle reference as an input and the output is the steering angle of a wheel. For tuning the linearized model was used. The model equations are used from [7].

The front tire slip angle $\alpha_F$ is defined as the angle between the orientation of the front wheel and the velocity vector at the front contact patch. Assuming small angles ($\sin\beta \approx \beta$, $\cos\beta \approx 1$) and constant longitudinal velocity $v \approx v_x$, the expression simplifies to:

$$\alpha_F \approx -\arctan\left(\beta + \frac{l_f}{v_x} r\right), \tag{6.5}$$

where $r = \dot{\psi}$. Linearizing further using $\arctan(x) \approx x$ for small $x$, we obtain:

$$\alpha_F \approx -\left(\beta + \frac{l_f}{v_x} r\right). \tag{6.6}$$

When front wheel steering input $\delta$ is included, the effective front slip angle becomes:

$$\boxed{\alpha_F = \delta - \beta - \frac{l_f}{v_x} r} \tag{6.7}$$

It shows that the slip angle is a linear combination of the steering angle $\delta$, the vehicle sideslip angle $\beta$, and the yaw rate $r$, scaled by the geometry and speed of the vehicle. In the state-space model, this output can be defined using:

$$y = Cx + D\delta, \quad \text{with} \quad C = \begin{bmatrix} -1 & -\frac{l_f}{v_x} \end{bmatrix}, \quad D = 1 \tag{6.8}$$

This allows the computation of the transfer function $G_{\alpha_F}(s) = \frac{\alpha_F(s)}{\delta(s)}$ by combining the linear vehicle dynamics with the slip angle output definition.

The rsulting transfer function is:

$$G_\alpha(s) = \frac{s^2 + 3.645s + 12.88}{s^2 + 6.961s + 11.06} \tag{6.9}$$

The system exhibits second-order dynamics in both the numerator and denominator, which introduces nontrivial zero dynamics that significantly influence the transient response. This structure suggests that the model captures both the resonant characteristics of the vehicle body and an active zero shaping effect, though it may not accurately reflect the expected low-pass behavior of a typical lateral dynamics response.

The PI controler constants were set to $K_p = 1$ and $K_i = 9$. Transfer function of the closed loop system is:

$$G_{\alpha\text{closed}}(s) = \frac{5s^2 + 60s + 135}{s^3 + 11s^2 + 71s + 135} \tag{6.10}$$

The closed-loop system is stable with one real pole at $-2.81$ and a pair of complex conjugate poles at $-4.09 \pm j5.59$, resulting in a moderately underdamped response characterized by a rise time of 0.198 s, settling time of 0.71 s, and an overshoot of approximately 12.1%.

To tune th outer loop the output in form of lateral acceleration is used. To get $a_y$ we bi-linear model of tire was used to linearize Pacejka model. The conversion can be done using:

$$a_y = \frac{F_y}{m} = \frac{C_\alpha F_{z_f} \cdot \alpha_F}{m} \tag{6.11}$$

where C is used from table 4.1. If we define:

$$K_a = \frac{C_\alpha F_{z_f}}{m} \tag{6.12}$$

we can write the transfer function from slip angle reference to lateral acceleration as:

$$G_{a_y}(s) = K_a \cdot PI(s) \cdot G_\alpha(s) \tag{6.13}$$

The values of the PI controller were set to $K_p = 0.4$ and $K_i = 3$. In figure 6.3 we can see the step response of the closed loop system.

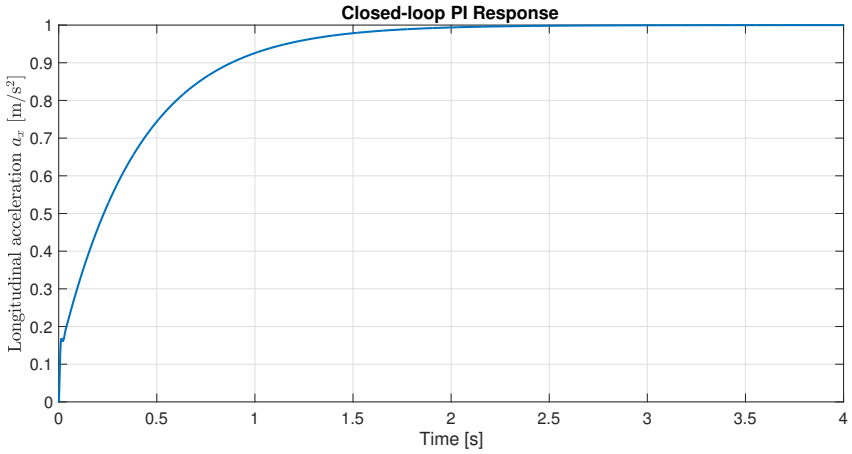



**Figure 6.3:** Step response of close loop.

## 6.4 Simulation results

As with the two-wheel steered platform, two different simulations were run. The first one was a step change of the reference signals. The same reference values were used as for the LQR controller. Running the same type of simulation allows for more close analysis between the two platforms. This will be discused in the following chapter. Here in figure 6.4, we can see the evolution of vehicle's state and input vector. Tracking the reference of yaw rate is done without any constant error. This however, cannot be said for the velocity vectors. We can see that the reference tracking of velocity in x has a constant error and the velocity drops with increased reference in lateral velocity. The vector of velocity in the lateral direction tracks the reference with constant error, however the error is smaller than the one for longitudinal velocity.

In the time interval between 30 and 40 seconds, the controller is supposed to settle the car back to normal driving. However, it finds unstable equilibrium. During normal driving, the wheel speeds on front and rear wheels are supposed to be very similar. When no acceleration and no steering is applied the most efficient way to drive is to have both wheel speeds equal. This behaviour can be seen during the first ten seconds of the experiment. In the time interval between 30 and 40 seconds however the controller overspins the front wheel and brakes the rear wheel. This is not efficient and it would be inadequate for implementation on any real vehicle. During the drift phases, the controller is able to maintain the high sideslip angle and yaw rate. As we can see, the four-wheel steering is being used with steering angle higher than the front, while both beeing around 30 degrees.

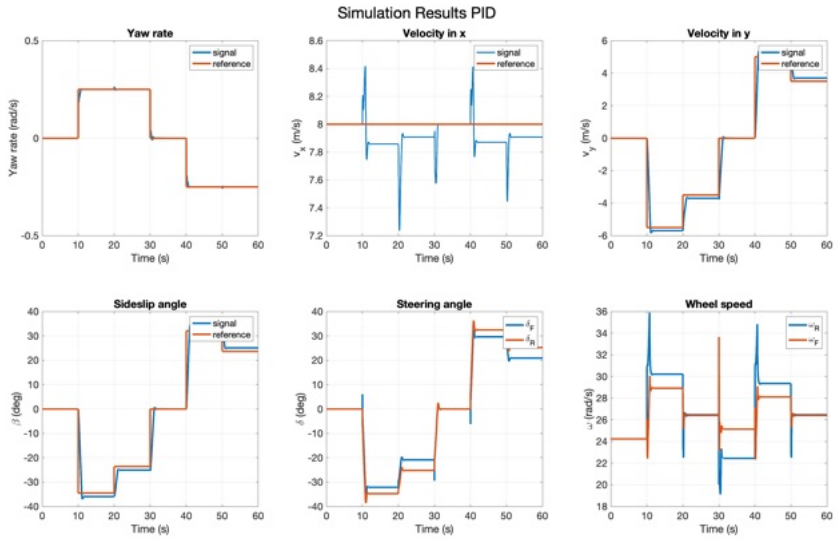



**Figure 6.4:** Vehicle state vector and input vector during simulation.

Figure 6.5 presents slip variables of the model. Here, during the initiation of the high sideslip angle maneuver, both the slip ratio and slip angle rise to the peak of the slip curve. After this brief period of time, both settle to around safe values. As in previous figure in the interval between 30 and 40 seconds, we can see rear slip ratio dropping below zero, thus indicating braking. On the other hand the front slip ratio stays above zero, indicating acceleration. Both previously presented figures indicate stabilization of the car during the maneuver.

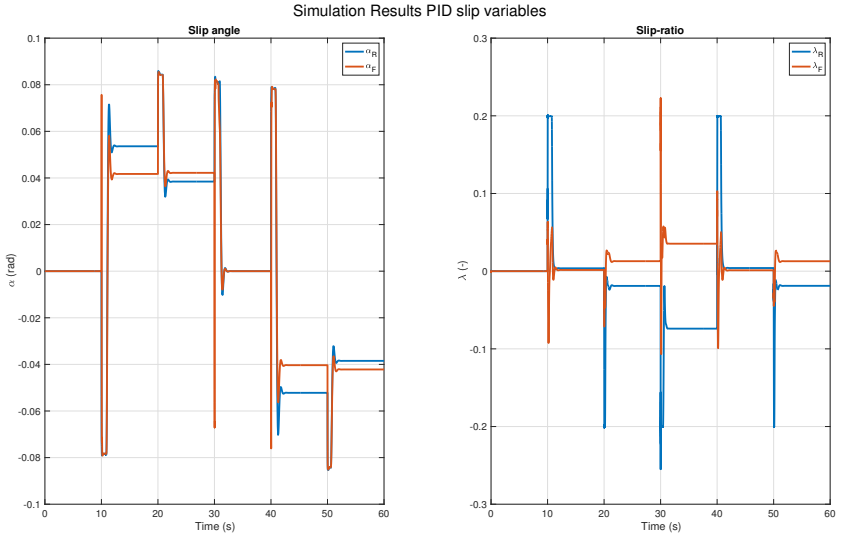



**Figure 6.5:** Rear weel slip variables simulation results.

From the graphed position displayed in figure 6.6 we canthat the car does not spin and the drift is stabilized. Slight deviation to the side opsote of the turning is present during the initialtion of the maneuver.

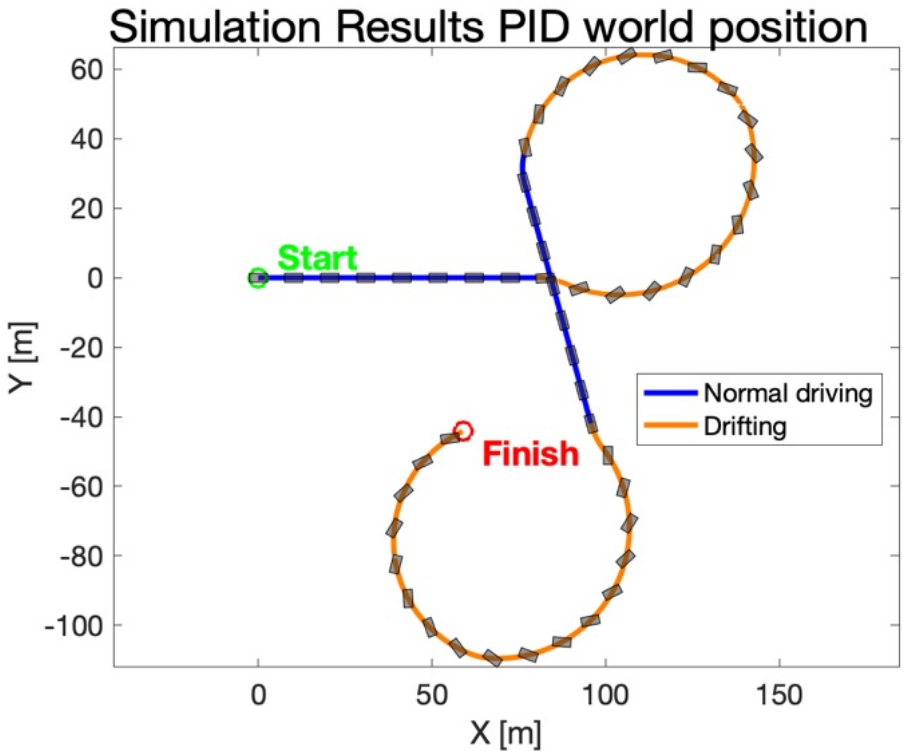



**Figure 6.6:** Car position in world frame during simulation.

The second simulation was done with ramp change of the $v_y$ reference whilst maintaing the yaw rate reference. In figure 6.7 we can see the car states. As in previous simulation the tracking of a yaw rate reference is withou a constant error. As in previous simulation we can see that with increase in reference value for the lateral velocity the longitudinal velocity is proportionaly droping thus increasing a tracking error. The Longitudinal velocity is being trcked without an error. During the ramp simulationt the sideslip angle of the car reached as much as 45 degrees whilst the car was stable in the maneuver.

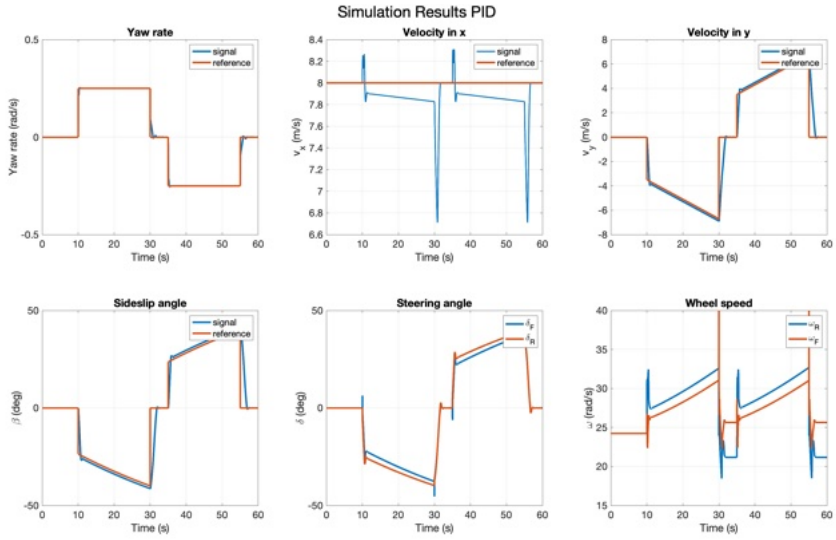



**Figure 6.7:** Vehicle state vector and input vector during simulation.

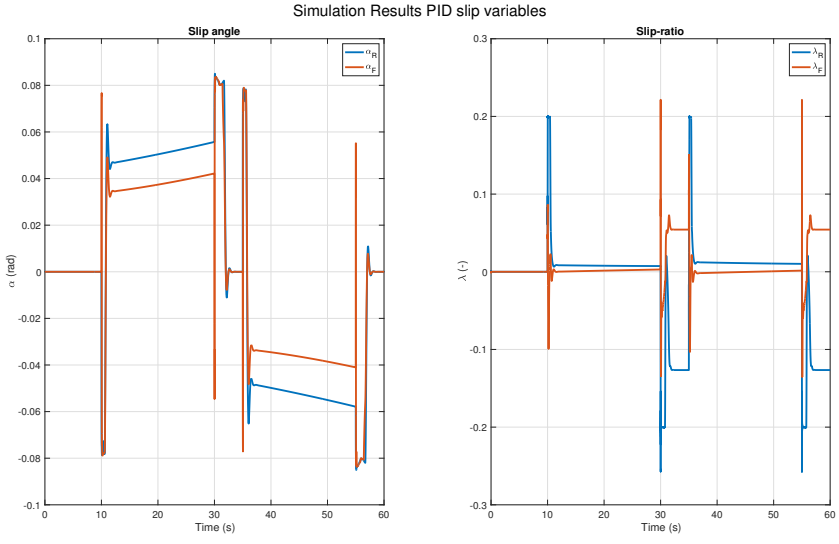



**Figure 6.8:** Rear weel slip variables simulation results.

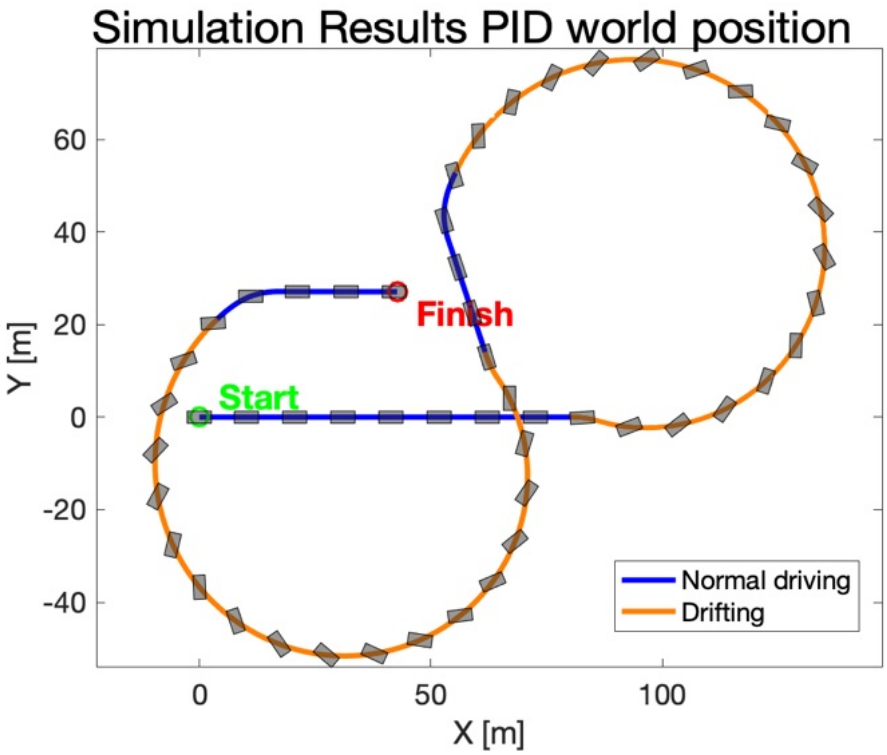



**Figure 6.9:** Car position in world frame during simulation.

# Chapter 7

# Platform comparison

This chapter presents a comparative analysis of simulation results. As presented in the beging of the thesis variable on whitch the performance of the controllers but mosty the platforms can be drawn and comparison can be made. Here the results will be compared based on gained slip, generated forces and velocities. Co

## 7.1 Slip Variables Comparison

The slip angle on the rear tire are shown in figure 7.1. Here we can see as the rear tire can not be s steered and reference value for the side-slip angle $\beta$ is well above 30 degrees. The slip angle on the front tire has to reflect that. Slip angle on the rear tire is the same as side-slip angle of the car.

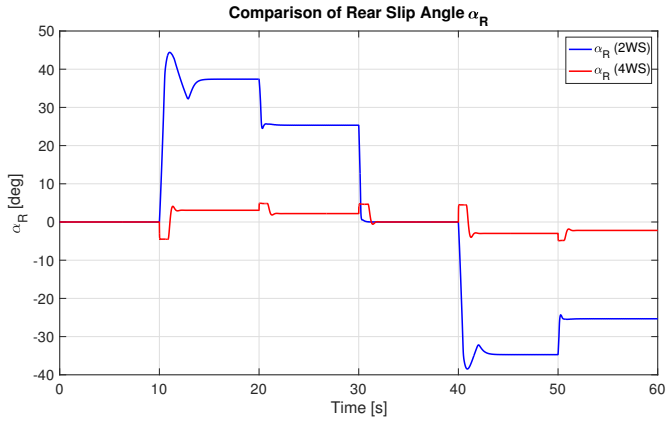



**Figure 7.1:** Comparison of rear wheel slip variables for various maneuvers.

In figure 7.2 we can see the slip ratio on the rear tire. The slip ratio for two wheel steer platform is well above the force saturatio level during the initiation of drift maneuver. When settled the value still remains very high but in the safe region this indicates that the tire are sliding only sideways and are not overspun. For the 4WS platform the slip ratio rises briefly during the drift initiation and than settles to a value form normal driving interval.

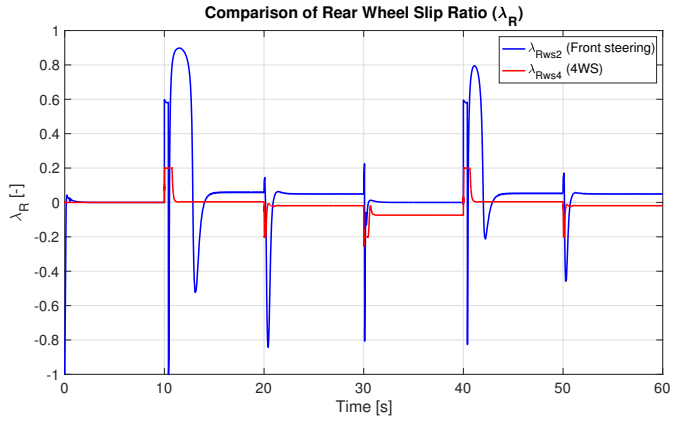



**Figure 7.2:** Comparison of rear wheel slip variables for various maneuvers.

## 7.2 Wheel Forces Comparison

The Figure 7.3 wheel Forces generated on the rear tire are shwown. We can see that forces needed to maintain the drift are much higher for the 2WS platform. We can also see initial spike for the 2WS platform during the drift initiation. This is due to the fact that the car has to move beyond linear region of the tire model. The 4WS platform has to generate much lower forces to maintain the car in the drift.

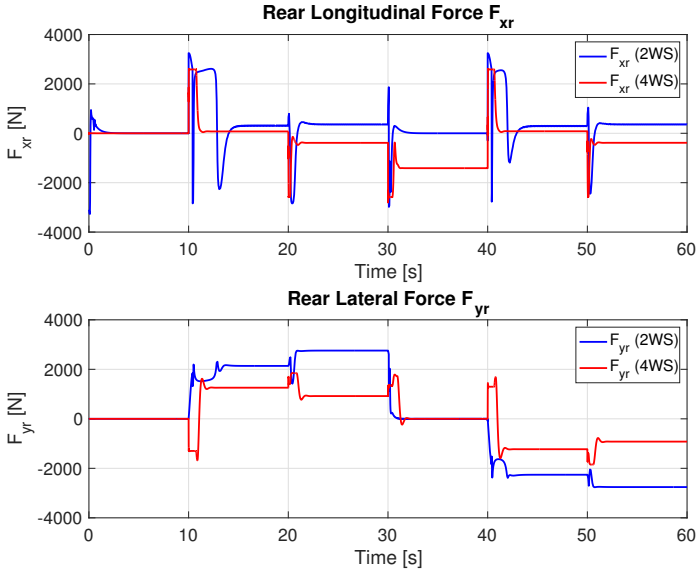



**Figure 7.3:** Comparison of wheel forces derived from the tire models across the simulation scenarios.

## 7.3 Mechanical Power Comparison

To compute a mechanical power a following equation was used on each platform:

$$P = |(F_{xr} + F_{xf}) \cdot v_x| + |(F_{yr} + F_{yf}) \cdot v_y| \tag{7.1}$$

As expected the 4WS platform has much lower Mechanical poower output during drift. This is due to the fact thath it has much lower slip variables. The 2WS platform has to generate much higher forces to maintain the car in the unstable zone. The power difference during the drift maneuver is the amount of power that is disapated in the sliding rear tire on 2WS platform.

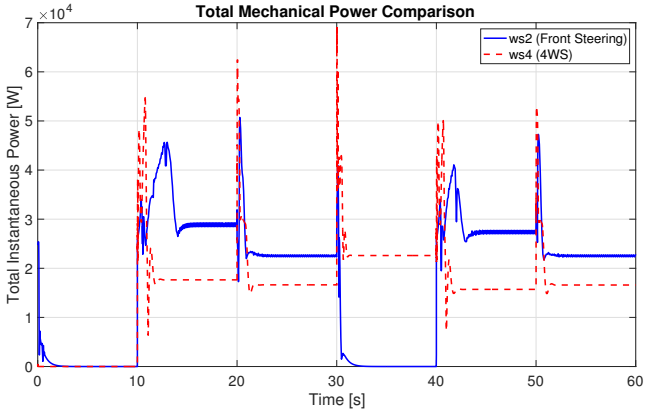



**Figure 7.4:** Comparison of the mechanical power outputs during the maneuvers.

# Chapter 8

# Conclusion

This thesis explored the problem of vehicle stabilization during high side-slip angle maneuvers, commonly observed in drifting and power-sliding scenarios. A nonlinear single-track model incorporating tire dynamics was implemented and used to evaluate the performance of two distinct vehicle architectures: a conventional two-wheel steering (2WS) platform and an overactuated four-wheel steering (4WS) platform.

Controllers were designed for both configurations. For the 2WS vehicle, a reference-tracking Linear Quadratic Regulator (LQR) was implemented in conjunction with a PI controller for longitudinal control. The controller was augmented with a drift initiator logic to enable autonomous transitions between normal driving and drifting modes. For the 4WS platform, a cascaded PID control architecture was used to independently manage lateral and longitudinal dynamics using control allocation strategies.

The conducted simulations included a variety of reference tracking scenarios. Results showed that:

- The 4WS platform achieved superior control authority during drifting, enabling smoother transitions and better tracking performance.
- Slip ratios and tire forces were significantly reduced in the 4WS configuration, indicating less aggressive tire utilization and improved efficiency.
- The 4WS platform demonstrated lower lap times and shorter racing lines, highlighting its maneuverability advantage.

- Mechanical power output was lower during drift maneuvers for the 4WS platform, suggesting reduced energy dissipation due to optimized force distribution.

Overall, the findings confirm that four-wheel steering can substantially improve high side-slip maneuver control, both in terms of stability and efficiency. The use of rear steering allows the vehicle to initiate and maintain drifts with lower lateral forces and slip values, potentially reducing tire wear and improving safety in high-performance or autonomous driving contexts.

The work could serve as a foundation for future research in real-world implementation of autonomous drifting controllers and vehicle control architectures with higher degrees of actuation freedom.

# Chapter 9

# Future work

- Create a more complex model of the vehicle dynamics

  Model presented in this thesis was a simplification. In the future a twintrack model with weights distribution would be a way to implement high fidelity simulation for verification of the controller.

- Create higher level path tracking controller.

  In the furture a path planning should be done to create an optimal trajectory optimizing lap time, path tracking algorithm is to be implemented as well. Created algorithm should in real time be able to decide of the best maneuver to be done ether clasic driving or high sideslip angle driving.

- Create a look-up table for LQR for different speeds

  As implemented in this thesis the LQR controller was designed for a longitudinal speed around 8m/s. In the future a look-up table for different speeds should be created. This would allow for a more precise controller design as well as a more precise controller in veariing conditions, speeds and yaw rates.

- Test and adapt 2WS controller on real platform

  The controller was designed and tested on a simulation model. In the future the controller should be tested on a real platform. This would allow for a more precise tuning of the controller as well as testing the robustness of the controller. As presented in the state of the art chapter there are resarch groups that alredy working and do have real world results.

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